CHARUSAT
CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

Faculty of Medical Sciences

Ashok & Rita Patel Institute of Physiotherapy (ARIP)

Master of Physiotherapy(MPT) (Paediatrics)

ACADEMIC REGULATIONS & SYLLABUS DETAILS

AY 2019-20

Ashok & Rita Patel
Institute of Physiotherapy

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

FACULTY OF MEDICAL SCIENCES

ASHOK & RITA PATEL INSTITUTE OF PHYSIOTHERAPY

M.P.T. PROGRAMME

COURSE REGULATIONS

1. OBJECTIVES:

The Post Graduate of Physiotherapy programme:

- Integrate knowledge of basic sciences and physiotherapy in order to modify treatment approachesthat reflect the breadth and scope of physiotherapy practice.
- Integrate the use of basic principles of research in critical analysis of concepts and findings generatedby self and others.
- Practice in an ethical and legal manner that values the professionalism.
- Prepare leaders in the multifaceted roles of clinicians, educators, researchers, and administrators inindividual, group, and community contexts model.
- Facilitate commitment to independent thinking and lifelong learning and realization of the intrinsicrewards of these attributes.
- Design a plan of care that synthesizes best available evidence and patient preferences, implementing safe and effective psychomotor interventions, and determining the efficacy of patient outcomes.
- Develops interests for care of the individual and provide physiotherapy services to the community.

2. ELIGIBILITY FOR ADMISSION:

Eligibility of a candidate for admission to Master of Physiotherapy programme will be according to the regulations for admission decided by CHARUSAT norms from time to time.

3. DURATION OF THE COURSE:

The duration of Master of Physiotherapy programme shall be of two academic years (4 semesters).

4. MEDIUM OF INSTRUCTION:

English shall be the medium of instruction for all the courses of study and for the examinations.

5. ATTENDANCE:

A candidate is required to attend atleast 80% of the total classes conducted in a semester in each course prescribed for that semester, separately in theory, practical and clinical practice.

6. COURSES OF STUDY:

The courses of M.P.T. programme of two years are given in **ANNEXURE I.**

7. COURSE EVALUATION:

7.1 The performance of every student in each course for university examination will be evaluated asfollows:

7.1.1 Internal evaluation by the course faculty member(s) based on continous assessment, for 30% ofthe marks for the course;

7.1.2 Final examination by the university through written paper or practical examination or oral examination by the student of combination of any two or more of these, for 70% of the marks of the course.

7.2 Internal evaluation:

7.2.1 The internal assessment is done based on continous evaluation method. Every semester, there will be two internal examinations for theory and practical. For the award of internal marks in theory and practical, the average of the two tests shall be considered along with other components like attendance, case presentations, microteaching, journal clubs and assignments.

7.3 University (External) Examination:

7.3.1 Every student has to score minimum 40 % of marks in theory and practical examination separately to pass in the final University Examination.

7.4 Aggregate:

7.4.1 Every student has to have an aggregate score of 50 % of marks of both in theory and practical examination separately to pass in the final University Examination and the grade will be awarded based on the aggregate marks.

8. GRADING:

8.1 The total of the internal evaluation marks and final University examination marks in each course willbe converted to a letter grade on a ten-point scale as per the following scheme:

Range of Marks (%)	**≥80**	**≥75 <80**	**≥70 <75**	**≥65 <70**	**≥60 <65**	**≥55 <60**	**≥50 <55**	**<50**
Letter Grade	**AA**	**AB**	**BB**	**BC**	**CC**	**CD**	**DD**	**FF**
Grade Point	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**0**

8.2 The student's performance in any semester will be assessed by the Semester Grade Point Average (SGPA). Similarly, his performance at the end of two or more consecutive semesters will be denoted by the Cumulative Grade Point Average (CGPA).

The SGPA and CGPA are defined as follows:

(i) SGPA = $\sum C_i G_i / \sum C_i$ where C_i is the number of credits of course i
G_i is the Grade Point for the course i and i = 1to n, n = number of courses in the semester

(ii) CGPA = $\sum C_i G_i / \sum C_i$ where C_i is the number of credits of course i
G_i is the Grade Point for the course i
and i = 1 to n, n = number of courses of all semesters up to which CGPA is computed.

(iii) No student will be allowed to move further if CGPA is less than 3 at the end of first year.

(iv) In addition to above, the student has to complete the required formalities as per the regulatorybodies.

9. Awards of Degree:

9.1 Every student of the programme who fulfils the following criteria will be eligible for the award ofthe degree:

9.1.1 He/She should have earned at least minimum required credits as prescribed in course structure,

9.1.2 He/She should have cleared all internal and external evaluation components in every course,

9.1.3 He/She should have secured a minimum CGPA of 5.0 at the end of the programme,

9.1.4 He/She should have completed the dissertation.

9.2 The student who fails to satisfy minimum requirement of CGPA will be allowed to improve the grades in the final semester so as to secure a minimum CGPA for award of degree. Only latest grade will be considered.

10. Award of Class:

The class awarded to a student in the programme is decided by the final CGPA as per the following scheme:

Distinction: $10.0 \geq \text{CGPA} \geq 7.5$

First class: $7.5 > \text{CGPA} \geq 6.0$

Second Class: $6.0 > \text{CGPA} \geq 5.0$

Pass class: $5.0 > \text{CGPA} \geq 4.5$

11. Transcript:

The transcript issued to the student at the time of leaving the University will contain a consolidated record of all the courses taken, credits earned, grades obtained, SGPA,CGPA, class obtained, etc.

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY (CHARUSAT)

FACULTY OF MEDICAL SCIENCES
ASHOK & RITA PATEL INSTITUTE OF PHYSIOTHERAPY

CHOICE BASED CREDIT SYSTEM

FOR

MASTER OF PHYSIOTHERAPY
(PAEDIATRICS)

CHOICE BASED CREDIT SYSTEM

With the aim of incorporating the various guidelines initiated by the University Grants Commission (UGC) to bring equality, efficiency and excellence in the Higher Education System, Choice Based Credit System (CBCS) has been adopted. CBCS offers wide range of choices to students in all semesters to choose the courses based on their aptitude and career objectives. It accelerates the teaching-learning process and provides flexibility to students to opt for the courses of their choice and / or undergo additional courses to strengthen their Knowledge, Skills and Attitude.

1. CBCS – Conceptual Definitions / Key Terms (Terminologies)

Types of Courses: The Programme Structure consist of 3 types of courses: Foundation courses, Core courses, Elective courses.

1.1. Foundation Course

These courses are offered by the institute in order to prepare students for studying courses to be offered at higher levels.

1.2. Core Courses

A Course which shall compulsorily be studied by a candidate to complete the requirements of a degree / diploma in a said programme of study is defined as a core course. Following core courses are incorporated in CBCS structure:

A. University Core courses(UC):

University core courses are compulsory courses which are offered across university and must be completed in order to meet the requirements of programme. Environmental science will be a compulsory University core for all Undergraduate Programmes.

B. Programme Core courses(PC):

Programme core courses are compulsory courses offered by respective programme owners, which must be completed in order to meet the requirements of programme.

1.3. Elective Courses

Generally, a course which can be chosen from a pool of courses and which may be very specific or specialised or advanced or supportive to the discipline of study or which provides an extended scope or which enables an exposure to some other discipline / domain or nurtures the candidate's proficiency / skill is called an elective course. Following elective courses are incorporated in CBCS structure:

A. University Elective Courses(UE):

The pool of elective courses offered across all faculties / programmes. As a general guideline, Programme should incorporate 2 University Electives of 2 credits each (total 4 credits).

B. Programme Elective Courses(PE):

The programme specific pool of elective courses offered by respective programme.

1.4. Medium of Instruction

The Medium of Instruction will be English.

Ashok & Rita Patel Institute of Physiotherapy

Vision

To become a leading institute in evidence based education and research by preparing physiotherapists having proficiency and ethical values

Mission

To create physiotherapists with contemporary knowledge, entrepreneurial skills and ethical values to serve society

Program Outcomes: Physiotherapy

PO1 Communication: Demonstrate effective communication and interpersonal skills, which are adapted to meet the needs of diverse individuals and groups

PO2 Diagnosis and plan of care: Develop physiotherapy diagnoses and an individualized plan of care for the management and prevention of movement dysfunction

PO3 Evidence based practice: Apply principles of critical thinking and clinical reasoning to evidence-based physiotherapy practice.

PO4 Teaching and learning principles: Apply learning and teaching principles in educational, clinical and community settings

PO5 Team Member: Participate as an effective intra- and inter-professional team member

PO6 Ethical and legal standards: Adhere to safe, ethical and legal standards of current practice (as identified by professional organizations, central and state law and accrediting bodies)

PO7 Professional Responsibility and commitment: Develop responsibility and commitment to the profession and society through life-long learning and involvement in activities beyond job responsibilities

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY (CHARUSAT)

FACULTY OF MEDICAL SCIENCES

ASHOK & RITA PATEL INSTITUTE OF PHYSIOTHERAPY

MASTER OF PHYSIOTHERAPY (M.P.T.)

BRANCH: PAEDIATRICS

I[ST] SEMESTER

Course Code	Course Title	Teaching Scheme				Examination Scheme				
		Contact Hrs.			Credit	Theory		Practical		Total
		Theory	Practical	Total		Internal	External	Internal	External	
HS703.01F	Foreign Language (French/ German)		2	2	2	-	-	30	70	**100**
HS704F	Academic Speaking									
PT795.01	Health & Physical Activity	-	2	2	2	-	-	30	70	**100**
PT775	Clinical Education I	-	16	16	8	-	-	60	140	**200**
PT793	Research Methodology & Biostatistics	4	2	6	5	30	70	30	70	**200**
PT794	Measurement in Physiotherapy	4	-	4	4	30	70	-	-	**100**
PT966	Physiotherapy Education	3	2	5	4	30	70	30	70	**200**
PT873	Research Ethics									
PT875	Imaging in Physiotherapy									
TOTAL		**11**	**24**	**35**	**25**	**90**	**210**	**180**	**420**	**900**

2[ND] SEMESTER

Course Code	Course Title	Teaching Scheme				Examination Scheme				
		Contact Hrs.			Credit	Theory		Practical		Total
		Theory	Practical	Total		Internal	External	Internal	External	
HS705 F	Academic Writing	-	2	2	2	-	-	30	70	**100**
PT796.01	Fitness & Nutrition	-	2	2	2	-	-	30	70	**100**
PT778	Clinical Education II	-	16	16	8	-	-	60	140	**200**
PT777	Basic Sciences	4	-	4	4	30	70	-	-	**100**
PT798	Exercise Physiology & Nutrition	5	-	5	5	30	70	-	-	**100**
PT868	Exercise Programming	3	2	5	4	30	70	30	70	**200**
PT963.01	Literature Review									
PT870	Neonatal Care									
TOTAL		**12**	**22**	**34**	**25**	**90**	**210**	**150**	**350**	**800**

3RD SEMESTER

Course Code	Course Title	Teaching Scheme				Examination Scheme				
		Contact Hrs.			Credit	Theory		Practical		Total
		Theory	Practical	Total		Internal	External	Internal	External	
PT933.01	Clinical Education III	-	10	10	5	-	-	30	70	**100**
PT934.01	Dissertation I	-	15	15	11	-	-	120	280	**400**
PT935.01	Physical & Functional Diagnosis for Paediatrics	3	2	5	4	30	70	30	70	**200**
PT936.01	Advanced Physiotherapeutics for Paediatrics	4	2	6	5	30	70	30	70	**200**
TOTAL		**7**	**29**	**36**	**25**	**60**	**140**	**210**	**490**	**900**

4TH SEMESTER

Course Code	Course Title	Teaching Scheme				Examination Scheme				
		Contact Hrs.			Credit	Theory		Practical		Total
		Theory	Practical	Total		Internal	External	Internal	External	
PT937.01	Clinical Education IV	-	10	10	5	-	-	30	70	**100**
PT938.01	Dissertation II	-	26	26	20	-	-	200	400	**600**
TOTAL		**-**	**36**	**36**	**25**	**-**	**-**	**230**	**470**	**700**

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF MANAGEMENT STUDIES
DEPARTMENT OF HUMANITIES AND SOCIAL SCIENCES
HS703.01 F LANGUAGES (French)

I. Credits and Schemes:

Sem	Course Code	Course Name	Credits	Teaching Scheme	Evaluation Scheme				
				Contact Hours/Week	Theory		Practical		Total
					Internal	External	Internal	External	
I	HS703.01	LANGUAGES (French)	02	02	--	--	30	70	100

II. Course Outline

Module No.	Title/Topic	Classroom Contact Hours
1	**Introduction to French Language** **Facts and figures about French Language** **Basic French Linguistics** * Alphabets * Accents * Liaison * Nasalization French Culture, Differ between French and English **Grammar :** • Subject Pronoun • Verbs : (être ,avoir, habiter,regarder,manger … “er” verb) • Form of address • Numbers (1 to 20) • Nouns and plurals of nouns • The expression : C’est ,Il y a **Presentation:** 1) Self Introduction 2) Question and answering **Dialogue**	**08**
2	**Grammar :** • Definite articles • Indefinite articles • Present tense o Positive Froms o Negative Forms • Numbers (21 to 100 , 100-1000) • Days ,Months ,Familly • Verbs: (aller,venir,finir,pouvoir,vouloir ….. “ir “ verb) **Social Links:** 1) My family & relations 2) Appointments 3) Gathering information from someone	**08**

	Dialogue	
3	**Grammar :** • Common Adjectives • Comparative Adjectives • Common Adverbs • Interrogative Forms • The expression : “On” • Directions , Countries, Nationalities, Seasons, Weather, • Professions • Verbs: (Prendre, Apprendre, Comprendre, faire ... “re “ verb) **Work , Study and Travel** 1) Job/ Profession 2) Ticket Reservation (At Bus/At Railway/At Airport) **Dialogue**	**08**
4	**Grammar :** 1) Common Prepositions 2) Common Conjunctions 3) Past Tense 4) Future Tense 5) Colors ,Shapes, Animals ,Vegetables, Fruits 6) Verbs: (“er”, “ir”,”re” etc...) **Food & Shopping** 1) Buy a vegetables and fruits 2) Any Conversation between Customer and Vendor (At Mall/ At Restaurant / At Market) **Dialogue**	**06**
	Total	**30**

III. Instruction Methods and Pedagogy

The course is based on pragmatic learning. Teaching will be facilitated by Reading Material, Discussion, Task-based learning, assignments and various interpersonal activities like group/pair work, independent and collaborative work, presentations, etc.

IV. Evaluation:

The students will be evaluated continuously in the form of their consistent performance. There is no theoretical evaluation. There is just practical evaluation. The evaluation (practical) is schemed as 30 marks for internal evaluation and 70 marks for external evaluation.

The students’ performance in the course will be evaluated on a continuous basis through the following components:

Internal Evaluation:

Sl. No.	Component	Number	Marks per incident	Total Marks
1	Assignment / Presentation/ Task	5	5	25
2	Attendance and Class Participation			05
			Total	**30**

External Evaluation

The University Practical Examination will be for 70 marks and will test the LSRW skills in the French Language.

Sl. No.	Component	Number	Marks per incidence	Total Marks
1	Term Work, Viva and Practical (LSRW)	-	70	70
			Total	**70**

Course Outcome (COs):

After completion of the course, the student would:

CO1	Have a basic overview of the French language and culture.
CO2	Develop a basic understanding of the usage of French language such as alphabets,numbers,months , colour etc.
CO3	Learn the basics of French Grammar and their usages
CO4	Learn to use basic language functions such as introducing oneself, answering basic Wh questions
CO5	be able to initiate conversations/dialogues in the French language confidently.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	2	-	-	-	-	-	2
CO2	3	-	-	-	-	-	-
CO3	3	-	-	-	-	-	-
CO4	3	-	-	-	-	-	-
CO5	3	-	-	-	2	-	2

Correlation levels 1, 2 or 3 are as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

Note: The reference/ reading material will be provided in consultation with the expert.

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF MANAGEMENT STUDIES
DEPARTMENT OF HUMANITIES AND SOCIAL SCIENCES

HS704 F: ACADEMIC SPEAKING (Sem-I)

I. Credits and Schemes:

Sem	Course Code	Course Name	Credits	Teaching Scheme	Evaluation Scheme				
				Contact Hours/Week	Theory		Practical		Total
					Internal	External	Internal	External	
1	HS704 A/B/C/D/E/F/G/H	Academic Speaking	02	02	--	--	30	70	100

II. Course Outline

Module No.	Title/Topic	Classroom Contact Hours
1	**Foundations of Advance Communication** • *Meaning and Definition of Advance Communication* • *Advance Communication in Digital, Social, Mobile World* • *Strategies for Advance Communication* • *Meaning and Concept of Academic Language* • *High Frequency Academic Vocabulary*	**04**
2	**Art of Conversation** • *Describing people, places and things* • *Expressing opinions* • *Making suggesting* • *Persuading someone* • *Interpreting and Summarizing*	**06**
3	***Science of Power Speaking*** • *Phonemes* • *Word Stress* • *Pronunciation* • *Intonation* • *Pause* • *Register* • *Fluency* • *Prosody* • *Lexical Range*	**06**
4	***Academic Speaking Application – Part I*** • *Art of Oratory* • *Formal Presentation* • *Speech Analysis – Decoding Best Speeches*	**08**

5	**Academic Speaking Application – Part II** • *Job Interview* • *Group Discussion* • *Meeting*	**06**
	Total	**30**

III. **Instruction Methods and Pedagogy**

The course is based on practical learning. Teaching will be facilitated by reading material, discussion, task-based learning, projects, assignments and various interpersonal activities like case studies, group work, independent and collaborative research, presentations etc.

IV. Evaluation

The students will be evaluated continuously in the form of their consistent performance throughout the semester. There is no theoretical evaluation. There is just practical evaluation. The evaluation (practical) is schemed as 30 marks for internal evaluation and 70 marks for external evaluation.

Internal Evaluation

The students' performance in the course will be evaluated on a continuous basis through the following components:

Sl. No.	Component	Number	Marks per incidence	Total Marks
1	I-Talk	1	10	25
2	Situational Speaking	1	05	
3	Case Study - Speech Analysis	2	10	
4	Attendance and Class Participation	-		05
			Total	**30**

External Evaluation

The University Practical Examination will be for 70 marks and will test the advance communication skills and academic speaking.

Sl. No.	Component	Number	Marks per incidence	Total Marks
1	Viva / Practical	-	70	70
			Total	**70**

Course Outcome (COs):

After completion of the course the student would:

CO1	Understand and demonstrate advance communication skills and academic speaking.
CO2	Demonstrate linguistic competence
CO3	Demonstrate performing ability at group discussion and personal interview.
CO4	Demonstrate the formal presentation skills.
CO5	Demonstrate ability to communicate in diverse situations

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3	-	-	-	-	-	-
CO2	3	-	-	-	-	-	-
CO3	3	-	-	-	3	-	2
CO4	3	-	-	-	-	-	-
CO5	3	-	-	-	-	-	-

Correlation levels 1, 2 or 3 are as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

V. Reference Books

- Headway Academic Skills - Level 1: Listening, Speaking and Study Skills Student's Book Paperback

VI. Reading

- **Unit 1:** Business communication Today (Thirteenth Edition) by Courtland L. Bovee, John V. Thill and Roshan Lal Raina
- **Unit 2:** Effective Speaking Skills by Terry O' Brien
- **Unit 2:** Speak Better Write Better by Norman Lewis
- **Unit 2:** Well Spoken: Teaching Speaking to All Students by Erik Palmer
- **Unit 3:** Let Us Hear Them Speak : Developing Speaking – Listening Skills in English by Jayshree Mohanraj (Publisher – Sage Publication)
- **Unit 4:** The craft of scientific presentations: Critical steps to succeed and critical errors to avoid. New York: Springer by Michael Alley
- **Unit 4:** Presentation Skills in English by Bob Dignen (Publisher: Orient Black Swan)

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF MEDICAL SCIENCES
ASHOK & RITA PATEL INSTITUTE OF PHYSIOTHERAPY
MPT (Paediatrics) - Semester I
PT795.01 HEALTH & PHYSICAL ACTIVITY

Credit Hours:

Hrs. / Wk.			Credits			Total Marks		Total
L	P	T	L	P	T	Theory	Practical	
-	2	2	-	2	2	-	100	100

A. Outline of The Course:

Sr. No.	Title of the unit	Minimum number of hours
1.	INTRODUCTION TO HEALTH	10
2.	PHYSICAL ACTIVITY	15
3.	INTRODUCTION TO YOGA	15

Total Hours (Theory): 00

Total Hours (Practical): 40

Total Hours: 40

B. Content of the course:

1	**INTRODUCTION TO HEALTH**	**10 hrs**	**25%**
1.1	What is health?		
1.2	Healthcare delivery system: Developing Countries, Developed Countries		
1.3	Human anatomy, physiology & physical fitness		
1.4	Basics of Nutrition		
1.5	Life style disorders – obesity & diabetes		
2	**PHYSICAL ACTIVITY**	**15 hrs**	**37.5%**
2.1	What are Physical activity, exercise, physical fitness and epidemiology?		
2.2	Measurement of Physical Activity in individuals		
2.3	Physical Activity – Theoretical Perspective: Self-determination, trans theoretical		
2.4	Physical Activity and mental health – Body image, depression, problem with exercise		
2.5	Barriers & Facilitators of Physical Activity		
3	**INTRODUCTION TO YOGA**	**15 hrs**	**37.5%**
3.1	What is yoga?		
3.2	Types of yoga		
3.3	Benefits of yoga to various body systems		
3.4	Asanas, Pranayam		
3.5	Yoga therapy for various back pain, asthma, stress, hypertension, diabetes		

Course outcomes (COs)

At the end of the course, the student will be able:

CO1	Describe interrelation between physical activity, its physiological effects and changes in human anatomy.
CO2	Understand the health-related benefits of physical activity and risks associated with physical inactivity
CO3	Realize the areas of nutrition, cardiovascular health, diseases related to physical activity, stress management, substance use and abuse, and sexually transmitted diseases
CO4	Describe correlation of physical activity and mental health
CO5	Describe yoga, different types yoga and its application in different health disorders

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	2	-	3	3	-	-	2
CO2	-	3	3	2	-	2	3
CO3	-	2	2	3	-	2	2
CO4	1	2	2	2	-	1	2
CO5	3	3	1	3	3	-	3

Recommended Study Materials:

❖ **Textbooks:**

1. ACSM"s "Health Related Physical Fitness Assessment Manual Lippincott Williams and Walkins USA, 2005.
2. Nilima Patel (2008) Yoga and Rehabilitation, Jaypee Publication, India

❖ **Reference Books:**

1. Biddle, S. J. H., & Mutrie, N. (2008). Psychology of physical activity. London: Routledge
2. B.C. Rai. Health Education and Hygiene Published by Prakashan Kendra
3. Puri. K.Chandra.S.S. (2005). Health and Physical Education. New Delhi: Surjeet Publications

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF MEDICAL SCIENCES
ASHOK & RITA PATEL INSTITUTE OF PHYSIOTHERAPY
MPT (Paediatrics) - Semester I

PT775 - CLINICAL EDUCATION I

Credit Hours

Teaching Scheme				Examination Scheme				
Contact Hrs.			Credit	Theory		Practical		Total
Theory	Practical	Total		Internal	External	Internal	External	
-	16	16	8	-	-	60	140	**200**

Total Hours: 320 Hours

A. Outline of the Course:

Sl. No.	Skill Sets
1	Communication and motivation skills for relatives, care givers and members of the multidisciplinary team
2	Communication skills with children of all ages
3	Procedural skills and Knowledge about common outcome measures in paediatrics
4	Fine and gross motor evaluation in children
5	Sensory and motor assessment in children
6	Primitive reflex testing
7	GMFM, GMFCS scoring and interpretation
8	Diagnostic skills and knowledge
9	Common investigative procedures in children
10	Interpreting medical, educational and psychological assessments
11	Assessment and Evaluation Skills to record prognosis or progression

12	Problem focused management skills (Balance, strength, function, posture, mobility, standing and walking)
13	Exercise, play & recreational activities to optimize function of child with special need
14	Standardized Guidelines, research findings and government guidelines in service provision
15	Evidence based practice to the development and improvement of clinical practice
16	Effective family and patient education.
17	Case presentation of physical findings, evaluation, and diagnosis, patient education and therapeutic plan
18	Working effectively as a member of the rehabilitation team (Orthotics, podiatrics, speech & language therapist etc.)
19	Recognize own practice/knowledge and to request assistance from other health professionals if needed
20	Knowledge of legislative acts and child protection laws, sensitive issues such as child abuse
21	Knowledge about the governmental policies and guidelines for people with disability
22	Knowledge of ethical and medical-legal consideration in the care of children with physical limitations
23	Care, compassion and commitment in care of children with special needs
24	Professional Commitment and Attitude Towards improving the quality of care

Clinical training comprises all of the formal and practical "real-life" learning experiences provided for students to apply knowledge and skills in the clinical environment. Experiences would include those of short and long duration (e.g., part-time, full-time) and those that provide a variety of learning experiences (e.g., rotations on different units within the same practice setting such as Musculoskeletal Physiotherapy OPD & IPD etc.) to include comprehensive care of patients. Each student will be under the supervision of a clinical supervisor at the clinical education site who directly instructs and supervises students during their clinical learning experiences. Clinical supervisors will evaluate each student twice during clinical training (mid-term and final) using a standard evaluation tool.

The student will be formally evaluated through **Objective Structured Clinical Examination (OSCE)** by the internal and external examiners.

B. Course Outcomes:

At the end of the course, the students will be able to

CO1	Apply principles of sensitive and appropriate communication when working with children, their parents / grandparents / care givers.
CO2	Demonstrate skills in pediatric history taking, complete review of systems and physical examination
CO3	Identify appropriate assessment tools for specific paediatric conditions & conduct systematic assessments,
CO4	Develop a family and patient centered evidence based patient education and management plan.
CO5	Demonstrate positive attitude, responsibility, compassion, empathy and respect toward children and families.
CO6	Demonstrate knowledge of ethical and medical-legal consideration in the care of children with physical limitations

C. Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3	-	-	-	-	2	3
CO2	-	3	3	-	-	-	-
CO3	-	3	3	-	-	-	-
CO4	1	-	-	3	-	2	-
CO5	-	-	-	-	3	-	3
CO6	-	-	-	-	-	2	3

D. 1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF MEDICAL SCIENCES
ASHOK & RITA PATEL INSTITUTE OF PHYSIOTHERAPY
MPT (Paediatrics) - Semester I
PT793 RESEARCH METHODOLOGY & BIOSTATISTICS

Credit Hours:

Hrs. / Wk.			Credits			Total Marks		Total
L	P	T	L	P	T	Theory	Practical	
4	2	6	4	1	5	100	100	200

A. Outline of the Course:

Sr. No.	Title of the unit	Minimum number of hours
1.	INTRODUCTION TO RESEARCH	05
2.	STUDY DESIGNS	10
3.	SAMPLING	10
4.	DATA COLLECTION	15
5.	BIOSTATISTICS	35
6.	PUBLISHING AND PRESENTING RESEARCH	05

Total hours (Theory): 80 Hrs.

Total hours (Practical): 40 Hrs.

Total hours: 120 Hrs.

A. Detailed Syllabus:

1	**INTRODUCTION TO RESEARCH**	**05 Hours**	**6%**
1.1	Definition and need for research		
1.2	History of research in physiotherapy		
1.3	Research problem and research question		
1.4	Variables and its types		
1.5	Literature search using electronic databases		
2	**STUDY DESIGNS**	**10 Hours**	**13%**
2.1	Bias in research: types, the ways in which it can threaten study validity, measures to minimize bias in research		
2.2	Quantitative and qualitative research designs		
2.3	Internal and external validity of research		
2.4	Commonly used quantitative research designs and their major strength and limitations: Descriptive, observational, analytical and experimental study designs		
2.5	Overview of qualitative research methods		
3	**SAMPLING**	**10 Hours**	**12%**
3.1	Populations and samples		
3.2	Sampling methods: Probability and non-probability sampling methods		
3.3	Assignment to groups		
3.4	Sample size calculation		
4	**DATA COLLECTION**	**15 Hours**	**19%**
4.1	Primary versus secondary data collection		
4.2	Measurement error and the factors contribute to measurement error		
4.3	Outcome measures: types, validity, reliability and its uses in research		
5	**BIOSTATISTICS**	**35 Hours**	**44%**
5.1	Data display and summary		
5.2	Summary statistics for quantitative data		
5.3	Statements of probability and confidence intervals		
5.4	Choosing appropriate statistics		
5.5	Interpreting results of statistics such as p-value, confidence interval		
5.6	P-values, power, type I and type II errors		
5.7	Difference between clinical and statistical significance		
5.8	Use of computer software: Excel, SPSS		

6	**PUBLISHING AND PRESENTING RESEARCH**	**05 Hours**	**6%**
6.1	Types of publication		
6.2	Peer review process		
6.3	Components of research manuscript: IMRAD format		
6.4	Introduction to the EQUATOR network reporting guidelines for various study design		
6.5	Guidelines for presenting research: platform presentation, poster presentation		

Course Outcomes (COs):

At the end of the course, the student will be able to

CO1	Develop understanding on various kinds of research, objectives of doing research, research process, research designs and sampling techniques.
CO2	Able to distinguish a purpose statement, a research question or hypothesis, and a research objective.
CO3	Identify and discuss the role and importance of research in the physiotherapy.
CO4	Differentiate between qualitative and quantitative research.
CO5	Have basic knowledge of data analysis-and hypothesis testing procedures
CO6	Prepare a research proposal

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	-	2	2	1	-	3	3
CO2	-	1	-	1	-	2	1
CO3	1	-	-	2	1	-	-
CO4	-	-	1	3	1	3	2
CO5	-	-	-	-	-	3	3
CO6	-	-	2	1	3	3	1

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation "-"

Recommended Study Materials:

- **Textbooks:**
    1. Rehabilitation research: Principles and applications. Elizabeth Domholdt (2005). 3rd ed., 592 pp., Elsevier Saunders, St Louis, MI, USA.
    2. Statistics at Square One. Michael J. Campbell, T. D. V. Swinscow. October 2009, ©2010, BMJ Books

- **Reference Books:**
  1. Research in Physical Therapy. Editor-Christopher E. Bork Author. Published by J.B. Lippincott Co. (1993).
  2. Greenhalgh T. How to read a paper: The basics of evidence-based medicine. John Wiley & Sons; 2014 Feb 26.

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF MEDICAL SCIENCES
ASHOK & RITA PATEL INSTITUTE OF PHYSIOTHERAPY
MPT (Paediatrics) - Semester I
PT794 MEASUREMENT IN PHYSIOTHERAPY

Credit Hours:

Hrs. / Wk.			Credits			Total Marks		Total
L	**P**	**T**	**L**	**P**	**T**	**Theory**	**Practical**	
4	**-**	**4**	**4**	**-**	**4**	**100**	**-**	**100**

A. Outline of the course:

Sr. No.	Title of the unit	Minimum number of hours
1.	MEASUREMENT IN CLINICAL PRACTICE	12
2.	VALIDITY	23
3	MEASUREMENT ERROR	15
4	COMMON MEASUREMENT TOOLS USED IN PHYSIOTHERAPY	30

Total hours (Theory): 80

Total hours (Practical): 00

Total hours: 80

B. Detailed Syllabus:

1	**MEASUREMENT IN CLINICAL PRACTICE**	**12 Hours**	**15%**
1.1	Need for standardized measurement in clinical practice		
1.2	Purpose of measurement in clinical practice: Screening, diagnosis, prognosis and treatment outcomes		
1.3	Levels of measurement: Nominal, ordinal, interval and ratio level		
1.4	WHO's ICF as a framework for measuring functioning and disability		
	Classification of measurement tools: physiologic measures and biomarkers, performance based measures and self-reports, norm referenced and criterion referenced tests, disease specific and generic measures		
2	**VALIDITY**	**23 Hours**	**29%**
2.1	Content validity		
2.2	Construct Validity including convergent, discriminant concurrent and criterion validity		
2.3	Responsiveness to change		
2.4	Methods to evaluate various types of validity of measurement tools including responsiveness to change (MCID)		
2.5	Interpretation of validity assessment		
3	**MEASUREMENT ERROR**	**15 Hours**	**19%**
3.1	Measurement error: definition and sources of error		
3.2	Strategies to eliminate measurement error		
3.3	Estimation of measurement error associated with a measurement tool		
3.4	Reliability (test-retest, inter-intra rater reliability)		
3.5	Interpreting measurement error: Standard error of measurement (SEM) & The Bland- Altman plot, minimum detectable change (MDC), Receiver Operating Characteristic (ROC) curve		
4	**COMMON MEASUREMENT TOOLS USED IN PHYSIOTHERAPY**	**30 Hours**	**38%**
	Validity, interpretability and clinical utility of measures of the following:		
4.1	Measures of muscle performance: strength, tone, flexibility		
4.2	Measures of joint functions		
4.3	Measures of functional status		
4.4	Measures of pain		
4.5	Measures of balance and gait		
4.6	Measures of cardiovascular functions		
4.7	Screening tests of cognitive functions, nerve, muscle and other soft tissue integrity		

Course Outcomes (COs):

At the end of the course the student will be able to:

CO1	Recognize the value and importance of using standardized measurement tools in routine clinical practice.
CO2	Communicate the problems and biases that may arise when assessing constructs in clinical practice
CO3	Identify and define different types of reliability, distinguishing among types of reliability and their unique insights into the assessment of outcomes
CO4	Define measurement error, its types and recognize its influence in interpreting measurement scores
CO5	Evaluate and interpret the validity and errors of measurement tools in making clinical decisions about patient care.

Course Articulation Matrix

	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	-	-	3	-	-	-	-
CO2	-	2	3	-	-	-	-
CO3	-	-	3	2	-	-	-
CO4	-	-	3	-	-	2	-
CO5	-	2	3	2	-	2	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation "-"

Recommended Study Materials:

- **Textbooks**

1. Rehabilitation research: Principles and applications. Elizabeth Domholdt (2005). 3rd ed., 592 pp., Elsevier Saunders, St Louis, MI, USA.

2. Finch E, Brooks D, Stratford P, Mayo N. Physical Rehabilitation Outcome Measures: A Guide to Enhanced Clinical Decision Making. Lippincott Williams and Wilkins; 2nd Revised edition

- **Reference Books**

1. Portney LG, Watkins MP, Foundations of clinical Research: Applications to practice. FA Davis & Company;2015.
2. Mokkink LB, Terwee CB, Patrick DL, Alonso J, Stratford PW, Knol DL, Bouter LM, De Vet HC. The COSMIN checklist for assessing the methodological quality of studies on measurement properties of health status measurement instruments: an international Delphi study. Quality of life research. 2010 May 1;19(4):539-49.

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF MEDICAL SCIENCES
ASHOK & RITA PATEL INSTITUTE OF PHYSIOTHERAPY

MPT (Paediatrics) - Semester I
PT966 PHYSIOTHERAPY EDUCATION

Credits Hours:

Hrs. / Wk.			Credits			Total Marks		Total
L	P	T	L	P	T	Theory	Practical	
3	2	5	3	1	4	100	100	200

A. Outline of the course:

Sr. No.	Title of the unit	Minimum number of hours
1.	TEACHING & LEARNING	10
2.	CURRICULUM DESIGNING	10
3.	PRINCIPLES & METHODS OF TEACHING	10
4.	MEASUREMENT AND EVALUATION	10
5.	GUIDANCE AND COUNSELING AND AWARENESS PROGRAMME	10
6.	PHYSIOTHERAPY PROFESSION AND STAFF ROLES	10

Total hours (Theory): 60 Hrs.

Total hours (Practical): 40 Hrs.

Total hours: 100 Hrs.

B. Detailed Syllabus:

1	**CONCEPT OF TEACHING AND LEARNING**	**10 Hours**	**10%**
1.1	Meaning and scope of Educational Psychology.		
1.2	Meaning and Relationship between teaching and learning.		
1.3	Learning Theories.		
1.4	Dynamics of behavior.		
1.5	Individual Meaning and concept.		
2	**CURRICULUM**	**10 Hours**	**10%**
2.1	Basis of curriculum formulation.		
2.2	Framing objectives for curriculum.		
2.3	Process of curriculum development and factors involved.		
2.4	Evaluation of curriculum differences.		
2.5	Curriculum planning – Integrated teaching, Problem based learning, Evidence based medicine.		
2.6	Skill development- Clinical skills, Communication skills, Counseling skills.		
3	**PRINCIPLES AND METHODS OF TEACHING**	**10 Hours**	**10%**
3.1	Bloom's taxonomy of instructional objectives.		
3.2	Writing instructional objectives in behavioral terms.		
3.3	Unit planning, Lesson planning.		
3.4	Lecture, Demonstration Discussion, Seminar, Assignment.		
3.5	Types of teaching aids.		
3.6	Principles of selection, preparation and use of audio-visual aides.		
4	**MEASUREMENT AND EVALUATION**	**10 Hours**	**10%**
4.1	Nature of educational measurement: meaning, process, types of tests.		
4.2	Construction of an achievement test and its analysis.		
4.3	Standardized test.		
4.4	Introduction of some standardized tools, important tests of intelligence.		
4.5	Aptitude and personality.		
4.6	Continuous and comprehensive evaluation.		
5	**GUIDENCE AND COUSELLING AWARENESS PROGRAMME**	**10 Hours**	**10%**
5.1	Meaning & concepts of guidance and counseling.		
5.2	Principles of guidance and counseling.		
5.3	Awareness and guidance to the common people about health and diseases.		
6	**PHYSIOTHERAPY PROFESSION AND STAFF ROLES**	**10 Hours**	**10%**
6.1	Physiotherapy: Definition and Development.		
6.2	Physiotherapy practice in India and their demands. Physiotherapy services in rural		

	and urban areas.
6.3	History taking, assessment, tests, Patient communication, documentation of findings, treatment organization and planning/execution for intervention.
6.4	Documentation of rehabilitation assessment and management using International
6.5	Standardized tests and scales used in various types of cases for assessment and interpretation in Physiotherapy practice.
6.6	Roles of Physiotherapy Director, Physiotherapy Supervisor, Physiotherapy Assistant, Physiotherapy, Occupational therapist, Home Health Aide and Volunteer

Course Outcomes (COs):

At the end of the course, the students will be able to

CO1	Interpret the concept of teaching & learning
CO2	Design a curriculum for basic and advanced courses
CO3	Deliver the class in an effective manner by engaging all
CO4	Apply the managerial skills and knowledge in physiotherapy profession
CO5	Justify various teaching learning tools for effective assessment practices
CO6	Create counselling and awareness programme for various groups of patients

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	-	-	2	2	-	-	-
CO2	1	-	1	1	-	-	-
CO3	1	2		-	2	-	-
CO4	2		-	1	-	1	2
CO5	-	1	3	2	1	-	-
CO6	1	-	-	-	2	2	3

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation "-"

Recommended Study Materials:

❖ **Textbooks:**

1. Developing a Pedagogy of Teacher education: Understanding teaching and learning about teaching.

❖ **Reference Books:**

1. Physical Therapy Administration & Management by Hickik Robert J

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF MEDICAL SCIENCES
ASHOK AND RITA PATEL INSTITUTE OF PHYSIOTHERAPY
MPT (Paediatrics) - Semester I
PT873 RESEARCH ETHICS

Credit Hours:

Hrs. / Wk.			Credits			Total Marks		Total
L	P	T	L	P	T	Theory	Practical	
4	-	4	4	-	4	100	-	100

A. Outline of the course:

Sr. No.	Title of the unit	Minimum number of hours
1.	HISTORICAL PERSPECTIVES AND GUIDELINES TO HUMAN RESEARCH ETHICS	03
2.	NATIONAL & GLOBAL GUIDELINES, CODES AND REGULATIONS	15
3.	RESPECT FOR PERSONS: INFORMED CONSENT AND ASSENT PROCESS, VULNERABLE POPULATIONS	15
4.	SCIENTIFIC INTEGRITY & DATA MANAGEMENT	10
5.	PUBLICATION, PEER REVIEW AND AUTHORSHIP	10
6.	CONFLICT OF INTEREST & COMMITMENT	07

Total hours (Theory): 60 Hrs.

Total Hours (Practical):00 Hrs.

Total Hours (Practical): 00 Hrs.

A. Detailed Syllabus:

1.	**HISTORY AND FRAMEWORK FOR ETHICAL CLINICAL RESEARCH**	**03 Hours**	**5 %**
1.1	Ethics of clinical research, Historical Perspectives on Healthcare Research using human participants		
2.	**NATIONAL & GLOBAL GUIDELINES, CODES AND REGULATIONS**	**15 Hours**	**25 %**
2.1 2.2	Belmonte report, Nuremberg Code of Medical ethics, Helsenki Declaration CIOMS/WHO guidelines for epidemiological studies ICMR Guidelines for Biomedical Research using human participants: Ethical Principles of Ethical Research viz. essentiality, voluntariness, informed consent and community agreement, non-exploitation, privacy and confidentiality, precaution and risk minimization, Accountability and transparency professional conduct, public domain, totality of responsibility		
3.	**RESPECT FOR PERSONS: INFORMED CONSENT AND ASSENT PROCESS, VULNERABLE POPULATINS**	**15 Hours**	**25 %**
3.1 3.2 3.3	Informed Consent Process, documentation of informed consent, waivers of informed consent, Assent and Permission for Children's Participation in research, compensation issues for research participation Defining and determining vulnerability in context of health care research, Beneficence & Justice,Vulnerability and cognitive impairment, Inclusion of non-competent subjects, Safeguards for vulnerable and non-competent subjects, Ethical Concerns for Vulnerable Population involving pregnant females, human fetuses, neonates, children, persons who are physically handicapped, mentally disabled, economically disadvantaged, educationally disadvantaged, Institutionalized		
4.	**SCIENTIFIC INTEGRITY & DATA MANAGEMENT**	**10 Hours**	**16.67 %**
4.1 4.2	Importance of scientific integrity in healthcare research, behaviors that affect scientific integrity, allegations of scientific misconduct, adverse events, scientific record, maintenance, responsibility of researchers for maintaining accuracy of scientific records, data retention Institutional Ethics Committee : Mandate, Role & composition of IEC in ensuring scientificintegrity, Proposal Review Process and decision making		
5.	**PUBLICATION, PEER REVIEW AND AUTHORSHIP**	**10 Hours**	**16.67 %**
5.1 5.2	Purpose of publication, replication of results of new findings into clinical practice, avoiding fragmentary publication Authorship Issues: ICMJE guidelines to authorship, roles and responsibilities of authors and contributors		

5.3	Peer review: Purpose of peer review, elements of peer review		
6.	**CONFLICT OF INTEREST & COMMITMENT**	**07 Hours**	**11.66 %**
6.1	Defining COI for human research participants including authors, contributors, editorial boardmembers, reporting COI		

Course Outcomes (COs):

CO1	Describe the history and importance ethics of human subjects protections
CO2	Identify research activities that involve human subjects ethics
CO3	Discover the risks a research project might pose to participants and understand how to minimize the risks posed by a research project
CO4	Describe additional protections needed for vulnerable populations
CO5	Describe appropriate procedures for recruiting research participants and obtaining informedconsent
CO6	Identify the different professional committees that monitor human subjects protections

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3	-	-	3	-	-	-
CO2	-	3	-	-	-	3	-
CO3	-	-	-	3	3	3	-
CO4	3	-	-	-	3	-	-
CO5	-	-	3	3	-	-	-
CO6	-	-	-	-	3	3	3

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

Recommended Study Materials:

❖ **Textbooks**

1. Emanuel, E. J., Grady, C. C., Crouch, R. A., Lie, R. K., Miller, F. G., &Wendler, D. D.(Eds.). (2011). The Oxford textbook of clinical research ethics. Oxford University Press.

❖ **Scientific Papers**

1. Walanj, A. S. (2014). Research ethics committees: need for harmonization at the nationallevel, the global and Indian perspective. Perspectives in clinical research, 5(2), 66.
2. Sanmukhani, J., &Tripathi, C. B. (2011). Ethics in clinical research: The Indianperspective. Indian journal of pharmaceutical sciences, 73(2), 125.

3. Bhutta, Z. A. (2002). Ethics in international health research: a perspective from thedeveloping world. Bulletin of the World Health Organization, 80(2), 114-120.

- **Reference Books**

1. Long, T., & Johnson, M. (2007). Research ethics in the real world: issues and solutions forhealth and social care. Elsevier Health Sciences.
2. McNamee, M. J., Olivier, S., & Wainwright, P. (2006). Research ethics in exercise, healthand sports sciences. Routledge.

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

FACULTY OF MEDICAL SCIENCES

ASHOK & RITA PATEL INSTITUTE OF PHYSIOTHERAPY

MPT (Paediatrics) - Semester I

PT875 IMAGING IN PHYSIOTHERAPY

Credit Hours:

Hrs. / Wk.			Credits			Total Marks		Total
L	P	T	L	P	T	Theory	Practical	
3	2	5	3	1	4	100	100	200

A. Outline of the Course:

Sr. No.	Title of the unit	Minimum number of hours
1.	INTRODUCTION TO IMAGING	12
2.	RADIOLOGIC EVALUATION, SEARCH PATTERNS AND DIAGNOSIS	8
3.	RADIOLOGIC EVALUATION OF MUSCULOSKELETAL DISORDERS	16
4.	RADIOLOGIC EVALUATION OF NEUROLOGICAL DISORDERS	12
5.	RADIOLOGIC EVALUATION OF CARDIORESPIRATORY DISORDERS	12

Total hours (Theory): 60 Hrs.

Total hours (Practical): 40 Hrs.

Total hours: 100 Hrs.

B. Detailed Syllabus:

1	**INTRODUCTION TO IMAGING**	**12 Hours**	**20%**
1.1 1.2 1.3	Introduction to: Radiology (X-Ray), Computer tomography (CT), Magnetic resonance imaging (MRI), Diagnostic ultrasound, Angiography, Doppler and Echocardiogram. Role of the various imaging modalities: strengths, limitations, relative cost, associated risks, ionizing radiations, Contrast media-related issues. Integration of Imaging Into Physiotherapy Practice.		
2	**RADIOLOGIC EVALUATION, SEARCH PATTERNS AND DIAGNOSIS**	**8 Hours**	**13.3%**
2.1 2.2	Search pattern: the ABCs of radiologic analysis. Understanding the image and the radiologic report.		
3	**RADIOLOGIC EVALUATION OF MUSCULOSKELETAL DISORDERS**	**16 Hours**	**26.6%**
3.1	Routine radiologic evaluation & Advance imaging evaluation of common spine, upper limb and lower limb disorders.		
4	**RADIOLOGIC EVALUATION OF NEUROLOGICAL DISORDERS**	**12 Hours**	**20%**
4.1	**Routine radiologic evaluation & Advance imaging evaluation of common neurological disorders.**		
5	**RADIOLOGIC EVALUATION OF CARDIORESPIRATORY DISORDERS**	**12 Hours**	**20%**
5.1	Routine radiologic evaluation & Advance imaging evaluation of common cardiorespiratory disorders.		

Course Outcomes (COs):

At the end of the course, the students will be able to

CO1	Understand the concept of imaging techniques used in musculoskeletal, neurological and cardiorespiratory disorders.
CO2	Interpret the imaging based on the evidence, ethical and legal standards.
CO3	Integrate imaging concept in clinical assessment and management
CO4	Practice effective inter-professional communication in rehabilitation team.
CO5	Criticize the imaging study and communicate effectively with various stakeholders.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	-	3	3	3	-	-	-
CO2	-	-	3	-	-	3	-
CO3	-	3	3	-	-	-	-
CO4	-	-	-	-	3	-	-
CO5	3	-	-	3	-	-	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

Recommended Study Materials:

❖ **Textbooks:**

1. Lynn N. Mc Kinnis, Fundamentals of Musculoskeletal Imaging, 4th Edition, Philladelphia, F.A. Davis, 2014.
2. Lawrence Boxt, Suhny Abbara, Cardiac Imaging: The Requisites, 4th edition, Elsevier.
3. Jeffrey P Kanne, Clinically Oriented Pulmonary Imaging (Respiratory Medicine).
4. Pina Sanelli, Pamela Schaefer, Laurie Loevner, Neuroimaging: The Essentials (Essentials Series), First Edition.

❖ **Reference Books:**

1. Lynn N McKinnis, Michael Mulligan, Musculoskeletal Imaging Handbook: A Guide for Primary Practitioners. Philladelphia, F.A. Davis, 2014.
2. Ramón Ribes, Joan C Vilanova, Learning Musculoskeletal Imaging. Springer Heidelberg; London, New York, 2010.
3. Marcelo F. Di Carli, Raymond Y. Kwong, Novel Techniques for Imaging the Heart: Cardiac MR and CT (American heart association).
4. Robert L Kilkins, James R Dexter, Albert J Heuer, Clinical assessment in respiratory care, 6th Edition by Mosby, Elsevier.

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF MANAGEMENT STUDIES
DEPARTMENT OF HUMANITIES AND SOCIAL SCIENCES
HS705 (F): ACADEMIC WRITING

I. Credits and Schemes:

<table>
<tr><th rowspan="3">Sem</th><th rowspan="3">Course Code</th><th rowspan="3">Course Name</th><th rowspan="3">Credits</th><th>Teaching Scheme</th><th colspan="5">Evaluation Scheme</th></tr>
<tr><th rowspan="2">Contact Hours/Week</th><th colspan="2">Theory</th><th colspan="2">Practical</th><th rowspan="2">Total</th></tr>
<tr><th>Internal</th><th>External</th><th>Internal</th><th>External</th></tr>
<tr><td>II</td><td>HS705</td><td>Academic Writing</td><td>02</td><td>02</td><td>--</td><td>--</td><td>30</td><td>70</td><td>100</td></tr>
</table>

II. Course Outline

Module No.	**Title / Topic**	**Classroom Contact Hours**
1	**Academic Writing and Research Process** • Introduction to Academic Writing • Academic Writing as a Part of Research • Types of Academic Writing • Features of Academic Writing • Importance of Good Academic Writing in various Academic Works	**05**
2	**Anatomy of Academic Writing** • Academic Vocabulary • Simple and Complex Sentences • Organizing Paragraphs • The Writing Process • Adopting Academic Writing Style	**05**
3	**Key Academic Skills** • Note – taking • Note – making • Paraphrasing • Summarizing	**05**
4	**Accuracy in Academic Writing** • Lexical Range • Academic Language and Structures • Elements of Writing • *Proof Reading, Editing, and Rewriting*	**05**
5	**Using and Citing Sources of Ideas** • Academic Texts and their Types • Intellectual Honesty in Academic Writing • Avoiding Plagiarism – Idea Theft • Degrees of Plagiarism • Types of Borrowing • Anatomy of Citations • *Common Citation Styles*	**05**

6	**Contemporary Practices in Academic Writing** • Analytical Essays • Graph / Table / Process Interpretation and Description • Writing Reports • Writing Research / Concept Papers	**05**
Total		**30**

III. Instruction Methods and Pedagogy

The course is based on practical learning. Teaching will be facilitated by reading material, discussion, task-based learning, projects, assignments and various interpersonal activities like writing, group work, independent and collaborative research, etc.

IV. Evaluation

The students will be evaluated continuously in the form of their consistent performance throughout the semester. There is no theoretical evaluation. There is just practical evaluation. The evaluation (practical) is schemed as 30 marks for internal evaluation and 70 marks for external evaluation.

Internal Evaluation

The students' performance in the course will be evaluated on a continuous basis through the following components:

Sl. No.	Component	Number	Marks per incidence	Total Marks
1	Paragraph Writing	1	3	03
2	Note-taking / Note-making	1	3	03
3	Paraphrasing / Summarizing	1	4	04
4	Essay Writing	1	5	05
5	Concept Paper Writing	1	10	10
5	Attendance and Class Participation			05
			Total	**30**

External Evaluation

The University Practical Examination will be for 70 marks and will test the professional communication skills and academic writing skills of the students.

Sl. No.	Component	Number	Marks per incidence	Total Marks
1	Viva / Practical /Quiz/ Project / Academic Writing	-	70	70
			Total	**70**

Course Outcome (COs):

After completion of the course, the student would:

CO1	have sound understanding of the concept and applications of academic writing
CO2	have acquired enough knowledge of academic writing style, strategy and approach
CO3	be able to demonstrate error free and effective academic writing
CO4	be able to demonstrate ability to work on project/report/paper writing
CO5	understand the concept of plagiarism and learn to use different citation styles as a part of referencing

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3	-	-	-	-	-	-
CO2	3	-	-	-	-	-	-
CO3	3	-	-	-	-	-	-
CO4	3	-	-	-	3	-	-
CO5	3	-	-	-	-	-	2

Correlation levels 1, 2 or 3 are as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

V. Reference Books / Reading

Essential Reading for Concepts

- Academic Writing for International Students, Routledge
- Academic Writing: A Guide for Management Students and Researchers. Monipally, M. M. & Pawar, B. S. Sage. 2010. New Delhi

Essential Reading for Activity and Teacher Resource

- ***Effective Academic Writing Level - 1,2,3,4 (Second Edition) By:*** Alice Savage, Patricia Mayer, Masoud Shafiei, Rhonda Liss, & Jason Davis; *Publisher: Oxford*

Additional Reading

- Writing Your Thesis (2nd Edition) by Paul Oliver, Sage
- Development Communication In Practice by Vilanilam V J, Sage
- Intercultural Communication by Mingsheng Li, Patel Fay, Sage
- www.owl.perdue.edu

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF MEDICAL SCIENCES
ASHOK AND RITA PATEL INSTITUTE OF PHYSIOTHERAPY
University Elective- Semester II
PT796.01 FITNESS & NUTRITION

CREDIT HOURS:

Hrs. / Wk.			Credits			Total Marks		Total
L	P	T	L	P	T	Theory	Practical	
-	2	2	-	2	2	-	100	100

A. OUTLINE OF THE COURSE:

Sr. No.	Title of the unit	Minimum number of hours
1.	PHYSICAL EDUCATION AND PHYSICAL FITNESS	10
2.	NUTRITION AND HEALTH	15
3.	SPORTS AND LIFE SKILLS EDUCATION	15

Total hours (Theory): 00

Total Hours (Practical):40

Total Hours: 40

B. DETAILED SYLLABUS:

1	**INTRODUCTION TO THE USE OF DRUGS IN SPORT**	**10 Hours**	**25%**
1.1.	Concept of Physical Education, Meaning, Definition, Aims and Objectives of Physical Education, Need and Importance of Physical Education		
1.2	Physical Education and its Relevance in Inter Disciplinary Context		
1.3	Physical Fitness Components: Type of Fitness, Health Related Physical Fitness, Performance Related Physical Fitness		
2	**NUTRITION AND HEALTH**	**15 Hours**	**37.5%**
2.1	Concept of Food and Nutrition, Balanced Diet, Food Pyramid Index		
2.2	Macro and Micronutrients: Types, functions and classification system		
2.3	Carbohydrates: Types, RDA data, Glycemic index, Sources of Fats, saturated, unsaturated fats, recommended intake, importance of fat in diet, fats in health and disease		
2.4	Protein: Types EAA, function, assessing quality of proteins, selecting incomplete Proteins, RDA sources.		
2.5	Vitamins And Minerals: Types, functions, sources, and minerals - calcium, Phosphorus, iron, magnesium, sodium, potassium, and chloride. Trace elements - Sources and functions.		
2.6	Determining Caloric Intake and Expenditure, Obesity, Causes and Preventing Measures – Role of Diet and Exercise, Importance of hydration in exercise		
3	**SPORTS AND LIFE SKILLS EDUCATION**	**15 Hours**	**37.5%**
3.1	Sports and Socialization		
3.2	Physical Activity and Sport – Emotional Adjustment and Wellbeing		
3.3	Substance Abuse among Youth – Preventive Measures and Remediation		
3.4	Yoga, Meditation and Relaxation		
3.5	Sports and Character Building		
3.6	Values in Sports		
3.7	Sports for World Peace and International Understanding		

Course Outcomes (COs):

At the end of the course, the students will be able to

CO1	To underline general concepts of physical education and physical fitness.
CO2	To Appraise the importance of exercise in the maintenance of health and fitness.
CO3	To promote and understand the value of sports for life skill development.
CO4	Introduce nutritional principles and application to improve overall health.
CO5	To apply fitness related components in fitness assessment.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	2	1	1	1	-	-	-
CO2	1	1	2	3	1	-	-
CO3	1	-	2	3	1	-	-
CO4	1	1	2	3	-	-	-
CO5	1	3	2	1	-	-	1

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation "-"

Recommended study material:

❖ **Textbooks**

1. ACSM"s "Health Related Physical Fitness Assessment Manual Lippincott Williams and Walkins USA, 2005.
2. Siedentop D (1994) Introduction to Physical Education and Sports (2nd ed.), California: Mayfield Publishing Company.
3. Corbin Charles Beetal C.A., (2004) Concepts of Fitness and Welfare Boston McGraw Hill

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF MEDICAL SCIENCES
ASHOK & RITA PATEL INSTITUTE OF PHYSIOTHERAPY
MPT (Paediatrics) - Semester II
PT778 - CLINICAL EDUCATION II

Credit Hours

Teaching Scheme				Examination Scheme				
Contact Hrs.			Credit	Theory		Practical		Total
Theory	Practical	Total		Internal	External	Internal	External	
-	16	16	8	-	-	60	140	**200**

Total Hours: 320 Hours

A. Outline of the course:

Sl. No.	Skill Sets
1	Communication and motivation skills for relatives, care givers and members of the multidisciplinary team
2	Communication skills in children of all ages
3	Procedural skills and Knowledge about common outcome measures in paediatrics
4	Fine and gross motor evaluation in children
5	Sensory and motor assessment in children
6	Primitive reflex testing
7	GMFM, GMFCS scoring and interpretation
8	Diagnostic skills and knowledge
9	Common investigative procedures in children
10	Mental and emotional support skills of children and families
11	Rational for the prescription of specialized device and recommend purchase
12	Interpreting medical, educational and psychological assessments
13	Assessment and Evaluation Skills to record prognosis or progression
14	Problem focused management skills (Balance, strength, function, posture, mobility, standing and walking)
15	Exercise, play & recreational activities to optimize function of child with special need
16	Standardized Guidelines, research finding and government guidelines in service provision
17	Evidence based practice to the development and improvement of clinical practice

18	Effective family and patient education.
19	Case presentation of physical findings, evaluation, and diagnosis, patient education and therapeutic plan
20	Working effectively as a member of the rehabilitation team (Orthotics, podiatrics, speech & language therapist)
21	Recognize own practice/knowledge and to request assistance from other health professionals if needed
22	Knowledge of legislative acts and child protection laws, sensitive issues such as child abuse
23	Knowledge about the governmental policies and guidelines for people with disability
24	Knowledge of ethical and medical-legal consideration in the care of children with physical limitations
25	Care, compassion and commitment in care of children with special needs
26	Professional Commitment and Attitude Towards improving the quality of care

Clinical training comprises all of the formal and practical "real-life" learning experiences provided for students to apply knowledge and skills in the clinical environment. Experiences would include those of short and long duration (e.g., part-time, full-time) and those that provide a variety of learning experiences (e.g., rotations on different units within the same practice setting such as Musculoskeletal Physiotherapy OPD & IPD etc.) to include comprehensive care of patients. Each student will be under the supervision of a clinical supervisor at the clinical education site who directly instructs and supervises students during their clinical learning experiences. Clinical supervisors will evaluate each student twice during clinical training (mid-term and final) using a standard evaluation tool.

The student will be formally evaluated through **Objective Structured Clinical Examination (OSCE)** by the internal and external examiners.

Course Outcomes (COs):

At the end of the course, the students will be able to

CO1	Demonstrate communication skills that facilitate empathetic relationships and effective collaborations with families, children and other health care professionals.
CO2	Demonstrate skills of history taking from parents, children and adolescents in more complex situations.
CO3	Demonstrate understanding of the elements of a comprehensive paediatric physiotherapy assessment for children
CO4	Organize case presentation to reflect details of physical findings, reasons for evaluation, and justify the diagnostic, patient education and therapeutic plan.
CO5	Conduct SMART goal setting with appropriate evidenced based interventions for a child in conjunction with their family.
CO6	Demonstrate commitment to achieving personal and professional excellence in managing children with disabilities.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3	-	-	-	-	2	3
CO2	-	3	3	-	-	-	-
CO3	-	3	3	-	-	-	-
CO4	1	-	-	3	-	2	-
CO5	-	-	-	-	3	-	3
CO6	-	-	-	-	-	2	3

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF MEDICAL SCIENCES
ASHOK & RITA PATEL INSTITUTE OF PHYSIOTHERAPY
MPT (Paediatrics) – Semester II

PT777 Basic Sciences

Credit Hours:

Hrs. / Wk.			Credits			Total Marks		Total
L	P	T	L	P	T	Theory	Practical	
4	-	4	4	-	4	100	-	100

A. Outline of the course:

Sr No.	Title of the unit	Minimum number of hours
1.	EMBRYOLOGICAL DEVELOPMENT	13
2.	SENSORIMOTOR DEVELOPMENT	14
3.	MOTOR CONTROL AND MOTOR LEARNING	14
4.	POSTURAL CONTROL	13
5.	COORDINATED MOVEMENT PATTERNS	13
6.	NUTRITION AND GROWTH	13

Total hours (Theory): 80 Hrs.

Total hours (Practical): 00 Hrs.

Total hours: 80 Hrs.

B. Detailed Syllabus:

1	**EMBRYOLOGICAL DEVELOPMENT**	**13 hrs**	**16%**
1.1	Fertilization		
1.2	Intrauterine development of musculoskeletal, neurological and cardiopulmonary systems		
1.3	Risk factors associated with the development		
1.4	Identification of the various congenital development anomalies.		
2	**SENSORIMOTOR DEVELOPMENT**	**14 hrs**	**18%**
2.1	Developmental theories		
2.2	Developmental sequence		
2.3	Factors affecting sensorimotor development		
2.4	Differentiating typical and atypical development		
3	**MOTOR CONTROL AND MOTOR LEARNING**	**14hrs**	**18%**
3.1	Relationship of motor control with motor development		
3.2	Relationship of motor learning with motor development		
3.3	Motor control theories and assumptions		
3.4	Factors affecting motor learning		
3.5	Principles of motor learning		
4	**POSTURAL CONTROL**	**13 hrs**	**16%**
4.1	Biomechanical and musculoskeletal components		
4.2	Motor components		
4.3	Development of postural control		
4.4	Aging and postural control		
4.5	Abnormal postural control		
4.6	Changes in gait and analysis		
5	**COORDINATED MOVEMENT PATTERNS**	**13hrs**	**16%**
5.1	Reflexes and reactions		
5.2	Attitudinal postural reflexes		
5.3	Balance reactions		
6.	**NUTRITION AND GROWTH**	**13 hrs**	**16%**
6.1	Normal feeding patterns		
6.2	Components of balanced diet		
6.3	Age-matched requirements and modifications		
6.4	Globally accepted reference values		

Course Outcomes (Cos):

At the end of the semester the student will be able to:

CO1	Identify and communicate the developmental stages to patients and recognize various developmental disorders.
CO2	Interpreting the correlation of motor control and motor learning & its assumptions on motor development
CO3	Illustrate the knowledge of postural control development and its changes in a lifespan and distinguish the typical and atypical development.
CO4	Demonstrate skills of examination of reflexes and reactions on patients and their influence on the development.
CO5	Apply knowledge of feeding patterns and recommended phasic(monthly) diet meals for growing children to effectively communicate/educate with the mothers

Course Articulation Matrix

	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3	3	2	-	-	-	-
CO2	-	2	3	-	-	-	-
CO3	-	2	2	-	-	-	-
CO4	2	3	3	-	1	-	-
CO5	3	3	3	-	2	-	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation "-"

Recommended Study Materials:

❖ **Textbooks:**

1. Neurological Rehabilitation: Umphred, Darcy, A.
2. Physical therapy for Children: Suzann K. Campbell
3. Pediatric physical therapy: Jan S. Tecklin

❖ **Reference books:**

1. A Guide to Care and Management of Very Low Birth weight infants – A team approach: Caryl J. Semmler.
2. Decision making in Pediatric Neurologic Physical therapy: Suzann K. Campbell.
3. Developmental disabilities in Infancy and Childhood: Arnold J. Capute& Pasquale J. Accardo.
4. Understanding the nature of Sensory Integration with diverse populations: Susanne Smith Roley.
5. Exploring the spectrum of Autism and Pervasive Developmental Disorders: Carolyn Murray-Slutsky.
6. Developmental Care of Newborns & Infants: Carole Kenner.

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF MEDICAL SCIENCES
ASHOK & RITA PATEL INSTITUTE OF PHYSIOTHERAPY
MPT (Paediatrics) - Semester II
PT798 EXERCISE PHYSIOLOGY & NUTRTION

Credit Hours:

Hrs. / Wk.			**Credits**			**Total Marks**		**Total**
L	**P**	**T**	**L**	**P**	**T**	**Theory**	**Practical**	
5	**-**	**5**	**5**	**-**	**5**	**100**	**-**	**100**

A. Outline of the course:

Sr. No.	**Title of the unit**	**Minimum number of hours**
1.	PRINCIPLES OF EXERCISE PHYSIOLOGY	10
2.	EXERCISE TESTING AND PLANNING	25
3.	EXERCISE PRESCRIPTION AND TRAINING	25
4.	ENERGY TRANSFER/ DELIVERY SYSTEM	10
5.	ENERGY EXPENDITURE	10
6.	MEASUREMENT AND ASSESSMENT	10
7.	COMPONENTS OF NUTRITION	10

Total hours (Theory): 100

Total hours (Practical): 00

Total hour: 100

B. DETAILED SYLLABUS:

1	**PRINCIPLES OF EXERCISE PHYSIOLOGY**	**10 Hours**	**10%**
1.1	Role of Aerobic and Anaerobic mechanism during exercises.		
1.2	Acute effects of High, Burst and Short duration exercises.		
1.3	Acute effect of Steady level exercise on following parameters – Blood flow, Heart rate, Blood Pressure, Pulse Rate, Respiration Rate, Acid Base Balance, Body Temperature, Fluid-Electrolyte Balance and Substrate Utilization.		
1.4	Aging and physiologic function, physical activity in the different types of population.		
1.5	Clinical exercise physiology for cancer, cardiovascular and pulmonary rehabilitation.		
2	**EXERCISE TESTING AND PLANNING**	**25 Hours**	**25%**
2.1	Definition of exercise testing, the need for exercise testing, clinical assessment of exercise tolerance, factors affecting exercise tolerance, diagnostic use of exercise testing, indications of exercise testing, exclusion criteria, Clinical values of exercise testing.		
2.2	Objective assessment of exercise- related symptoms. Muscles fatigue and weakness, dyspnea, exertional chest pain, Applications in cardiac disorders, coronary artery disease, congenital heart disease, valvular disease, pulmonary vascular disease, peripheral vascular		
2.3	disease.		
2.4	Low Level Exercise Testing: Purpose, Contra - indications, Termination points.		
2.5	Maximal Exercise Testing: Purpose, Guidelines, Exercise test protocols, Contraindications and Precautions, Criteria for termination of test, Prognostic implications from exercise testing.		
2.6	Exercise electrocardiogram. The normal ECG. Normal response disease, changes associated with coronary disease. Changes in the ST segment and T waves, Effects of drugs, cardiac arrhythmias in exercise, abnormalities of conduction, the ECG in athletes.		
2.7	Approaches to clinical exercise testing, master step test, the balke protocol, the Bruce protocol, Scandinavian protocol, triangular protocol, walking protocol, Wingate test, maximal oxygen uptake, the stage I test, stage 2,3 and 4 tests, indications and contraindications to exercise testing.		
	Exercise tolerance test or stress test METS and their use in evaluation.		
3	**EXERCISE PRESCRIPTION & TRAINING**	**25 Hours**	**25%**
3.1	Introduction to exercise prescription, the individual approach, the aerobic session, frequency, time, mode of exercise, rate of progression, musculoskeletal conditioning, static stretching, systems of muscular strength and endurance training, Recommendation based on maximal exercise test results, interpretation of maximal exercise test results.		
3.2	Conditioning effects of various levels of Sub-Maximal Exercises. Considerations of age and sex in exercise and training.		

3.3 3.4 3.5	Exercise prescription for health and fitness with special emphasis to cardiovascular disease, Obesity and Diabetes, Fatigue – Types, Relevance with Exercise Tolerance tests & Training. Principles of health promotion for Growing Children, Healthy Adults, Pregnant/Lactating females, Elderly, Sports person. Special aids to exercise training and performance.		
4	**ENERGY TRANSFER/DELIVERY SYSTEM**	**10 Hours**	**10%**
4.1 4.2 4.2.1 4.2.2 4.2.3 4.2.4	Introduction to energy transfer, energy transfer in the body-phosphate bond energy, energy released from food, energy transfer in exercise. Systems of energy delivered and utilization: Cardiovascular system: cardiovascular regulation, integration and functional capacity. Pulmonary System: Dynamics of pulmonary ventilation, regulation of pulmonary ventilation, pulmonary ventilation during exercise, acid-base regulation. Endocrine system: organization, acute and chronic response to exercise. Role of Musculoskeletal system & Nervous System		
5	**ENERGY EXPENDITURE**	**10 Hours**	**10%**
5.1 5.2 5.3	Expenditure during rest, confinement during illness and various levels of Physical Exercises, factors influencing energy uptake and substrate utilization. Measurement of Human energy expenditure, individual differences and measurement of energy capacities. Energy expenditure during walking, jogging, running and swimming.		
6	**MEASUREMENT AND ASSESSMENT**	**10 Hours**	**10%**
6.1 6.2	Body composition assessment, physique, performance, and physical activity, over-weight, obesity and weight control. Lab Investigations - blood glucose, lipid profile, electrolytes, hemoglobin.		
7	**COMPONENTS OF NUTRITION**	**10 Hours**	**10%**
7.1 7.2 7.3 7.4 7.5	Carbohydrates: Types, RDA data, Glycemic index, Sources of Fats, saturated, Unsaturated fats, recommended intake, importance of fat in diet, fats in health and disease. Protein: Types EAA, function, assessing quality of proteins, selecting incomplete proteins, RDA sources. Vitamins And Minerals: Types, functions, sources, and minerals- calcium, Phosphorus, iron, magnesium, sodium, potassium and chloride. Trace elements- Sources and functions. Nutritional status in India, RDA by ICMR, basic five food groups, Principles of Menu planning, Nutritional deficiencies-protein calorie malnutrition, effects of Nutrition on teeth, role of fluoride. Diet–for Growing Age, Pregnancy, Lactation, Acute Illness, Convalescent Period, High level of Physical Activity, Aging & Sports, Obesity–prescription of Diet & its modification.		

7.6	Role of nutrient inhibitors. Effect of Food toxins and non-nutritive foods. Food Contamination and spread of diseases.
7.7	Types of diet–veg/non-veg. Advantages and disadvantages of the same, Food Processing, methods of food preparation.

Course Outcomes (COs):

At the end of the course, the students will be able to

CO1	Understand exercise physiology in healthy and diseased population
CO2	Understand body composition measurement, exercise and related symptoms testing in healthy and pathological conditions
CO3	Understand exercise prescription for healthy and various common diseased population
CO4	Understand components and types of diet for various population

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	-	-	-	3	-	-	-
CO2	-	3	3	-	-	-	-
CO3	-	3	3	3	-	-	-
CO4	-	-	3	3	-	-	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation "-"

Recommended Study Materials:

- ❖ **Textbooks:**
  1. Katch: Exercise physiology, energy nutrition and human performance.
  2. Scott K Powers: Theory and application to fitness and performance.
- ❖ **Reference Books:**
  1. Axen: Illustrated principles of exercise physiology.
  2. Frank: Exercise physiology for health care professionals.
  3. Tudor Hale: Exercise physiology – a thematic approach.
  4. George Brooks: Exercise physiology -Human bioenergetics and its application.

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF MEDICAL SCIENCES
ASHOK & RITA PATEL INSTITUTE OF PHYSIOTHERAPY
MPT (Paediatrics) - Semester II
PT963.01 LITERATURE REVIEW

Credit Hours:

Hrs. / Wk.			Credits			Total Marks		Total
L	P	T	L	P	T	Theory	Practical	
3	2	5	3	1	4	100	100	200

A. Outline of the Course:

Sr. No.	Title of the unit	Minimum number of hours
1.	INTRODUCTION TO LITERATURE REVIEW	10
2.	ACCESSING LITERATURE	15
3	SUMMARISING AND SYNTHESISING INFORMATION	15
4	WRITING UP THELITERATURE REVIEW	20

Total hours (Theory) : 60 Hrs.
Total hours (Practical): 40 Hrs.
Total hours: 100 Hrs.

B. Detailed Syllabus:

1	INTRODUCTION TO LITERATURE REVIEW	10 Hours	17%
1.1	Introduction and Definition		
1.2	Annotated bibliography vs. literature review		
1.3	Purpose of a literature review		
1.4	Types : narrative review and systematic review		
2	**ACCESSING LITERATURE**	**15 Hours**	**25%**
2.1	Formulating the search question		
2.2	Developing key words and search strategy		
2.3	PubMed Search: Using MeSH terms, advance search, use of limits and filters		
3	**SUMMARISING AND SYNTHESISING INFORMATION**	**15 Hours**	**25%**
3.1	Critical appraisal of the literature		
3.2	Structuring the literature review: The funnel method		
3.3	Coding and creating a matrix		
3.4	Analyzing the literature: Meta-analysis, Meta ethnography and thematic analysis		
4	**WRITING UP THE LITERATURE REVIEW**	**20 Hours**	**33%**
4.1	Sources of bias in creating and completing a literature review.		
4.2	Mistakes commonly made in reviewing research literature		
4.3	Criteria for good literature review		
4.4	Avoiding plagiarism and misinterpretation		

Course Outcome (COs):

At the end of the course, the student will be able to

CO1	Formulate an appropriate review question
CO2	Develop literature search strategy relevant to the review question
CO3	To critically appraise literature in the context of its contribution to understanding the research problem being studied.
CO4	Demonstrate how to construct and write a literature review

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	-	1	2	1	-	2	3
CO2	-	1	3	-	1	2	1
CO3	1	1	2	1	3	2	2
CO4	-	-	1	2	2	3	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

Recommended Study Materials:

- **Textbooks:**
  1. AveyardH. Doing a Literature Review in Healthand Social Care: A practical guide (3rd ed). Open University Press; 2014
- **Reference Books:**
  1. Ridley D. The literature review: A step-by-step guide for students. Sage Publications; 2012 Jul 23.
  2. Booth A . Systematicapproaches to a successful literature review. Sage Publications; 2012.

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF MEDICAL SCIENCES
ASHOK & RITA PATEL INSTITUTE OF PHYSIOTHERAPY
MPT (Paediatrics) - Semester II
PT870 – NEONATAL CARE

Credit Hours:

Hrs. / Wk.			Credits			Total Marks		Total
L	P	T	L	P	T	Theory	Practical	
3	2	5	3	1	4	100	100	200

A. Outline of the course:

Sr. No.	Title of the unit	Minimum number of hours
1.	INTENSIVE CARE OF NEONATE	11
2.	MEDICO-SURGICAL INVESTIGATIONS & PROCEDURES	15
3	ASSESSMENT	8
4	HIGH DEPENDENCY CARE OF NEONATE	8
5.	INFECTIONS IN NEONATES	10
6.	MANAGEMENT OF MORBIDITIES IN NEONATES	20
7.	FAMILY CENTERED CARE	8

Total hours (Theory): 48 Hrs.

Total hours (Practical): 32 Hrs.

Total hours: 80 Hrs.

B. Detailed Syllabus:

1	**INTENSIVE CARE OF NEONATE**	**11 Hours**	**15%**
1.1	Environment of NICU		
1.2	Early Intervention Services		
1.3	Potential complications of the NICU		
1.4	NICU experience and its relationship to Sensory Integration		
1.5	Transport of Neonates		
2	**MEDICO-SURGICAL INVESTIGATIONS & PROCEDURES**	**15 Hours**	**18%**
2.1	Resuscitation & Intubation		
2.2	Lumbar puncture, suprapubic tap, Chest tube placement, Genetic testing, Cerebral Function Monitor,		
2.3	Sonography, Urine tests		
2.4	Laser therapy, Phototherapy, ECMO		
2.5	Repair of esophageal atresia, gastroschisis, omphalocele, congenital diaphragmatic hernia, colostomy		
3	**ASSESSMENT**	**8 Hours**	**10%**
3.1	Developmental assessment		
3.2	Preterm & Prematurity developmental assessment		
3.3	Nutrition and feeding screening		
4	**HIGH DEPENDENCY CARE OF NEONATE**	**8 Hours**	**10%**
4.1	Problems of prematurity		
4.2	Theoretical perspectives for developmental supportive care		
4.3	Pain Management		
4.4	Palliative Care		
5	**INFECTIONS IN NEONATES**	**10 Hours**	**12%**
5.1	Timing of Infections		
5.2	Types of Neonatal infection		
5.3	Identifying and Treating Minor & Major Infections		
6	**MANAGEMENT OF MORBIDITIES IN NEONATES**	**20 Hours**	**25%**
6.1	Respiratory (RDS, ;BPD, MAS, Apnea of Prematurity, Birth Asphyxia, TTN)		
6.2	Neurological (PVL, HIE, IVH, Encephalopathy, NEC, ROP)		
6.3	Polycythemia, Anemia, Hypotension, Hypoglycemia, IEM, PDA		
7	**FAMILY CENTERED CARE**	**8 Hours**	**10%**
7.1	Professional- Parent Partnerships		
7.2	Parents’ responses and Parent-Infant Interaction		

7.3	Caregiving and the Environment
7.4	Sleep position protocols

Course Outcomes (COs):

At the end of the course, the students will be able to

CO1	Demonstrate skills of communication, assessment and management and recognize potential adverse outcomes for the neonate
CO2	Interpret indications, limitations, and gestational-age norms laboratory and imaging studies unique to the NICU setting
CO3	Evaluate, formulate a differential diagnosis with appropriate prioritization and manage common signs and symptoms of disease in high risk newborns
CO4	Demonstrate effective approach to assessment and daily management of seriously ill neonates using decision-making and problem solving skills
CO5	Demonstrate effectively as a part of an interdisciplinary team member in the NICU
CO6	Demonstrate knowledge of ethical and medical-legal consideration in the care of critically ill newborns

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3	3	-	-	-	2	1
CO2	-	3	2	-	-	-	-
CO3	-	-	3	2	-	-	-
CO4	-	-	2	3	-	-	-
CO5	2	-	-	-	3	-	-
CO6	-	-	-	-	-	2	3

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

Recommended Study Materials:

❖ **Textbooks:**

1. A Guide to Care and Management of Very Low Birth weight infants – A team approach, Caryl J. Semmler
2. Developmental Care of Newborns & Infants : Carole Kenner, Jacqueline M. McGrath
3. Pediatric physical therapy: Jan S. Tecklin

❖ **Reference Books:**

1.Developmental disabilities in Infancy and Childhood: Arnold J. Capute & Pasquale J. Accardo

2.Neurologic interventions for Physical therapy: Suzanne Martin

3.Neurological Rehabilitation: Umphred, Darcy, A.

4.Physical therapy for Children: Suzann K. Campbell

5.Decision making in Pediatric Neurologic Physical therapy: Suzann K. Campbell

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF MEDICAL SCIENCES
ASHOK & RITA PATEL INSTITUTE OF PHYSIOTHERAPY
MPT (Paediatrics) - Semester III
PT933.01- CLINICAL EDUCATION III

Credit Hours

Teaching Scheme				Examination Scheme				
Contact Hrs.			Credit	Theory		Practical		Total
Theory	Practical	Total		Internal	External	Internal	External	
-	10	10	5	-	-	30	70	**100**

Total Hours: 200 Hours

A. Outline of the course:

Sl. No.	Skill Sets
1	Communication and motivation skills for relatives, care givers and members of the multidisciplinary team
2	Communication skills in children of all ages with a range of complex conditions/disabilities to work with treatment programs
3	Parental education and communication skills
4	Procedural skills and Knowledge about common outcome measures in paediatrics
5	Fine and gross motor evaluation in children
6	Sensory and motor assessment in children
7	Primitive reflex testing
8	GMFM, GMFCS scoring and interpretation
9	Diagnostic skills and knowledge
10	Common investigative procedures in children
11	Mental and emotional support skills of children and families
12	Rational for the prescription of specialized device and recommend purchase
13	Evaluation for the need of home environment and School modification

14	Interpreting medical, educational and psychological assessments
15	Assessment and Evaluation Skills to record prognosis or progression
16	Problem focused management skills (Balance, strength, function, posture, mobility, standing and walking)
17	Exercise, Play and recreational activities to optimize the function of child with special needs
18	Standardized Guidelines, research finding and government guidelines in service provision
19	Evidence based practice to the development and improvement of clinical practice
20	Effective family and patient education.
21	Education for exercise adherence and home compliance
22	Case presentation of physical findings, evaluation, and diagnosis, patient education and therapeutic plan
23	Working effectively as a member of the rehabilitation team (Orthotics, podiatrics, speech & language therapist)
24	Evaluation for the need of referral services
25	Recognize own practice/knowledge and to request assistance from other health professionals if needed
26	Knowledge of legislative acts and child protection laws, sensitive issues such as child abuse
27	Knowledge about the governmental policies and guidelines for people with disability
28	Knowledge of ethical and medical-legal consideration in the care of children with physical limitations
29	Care, compassion and commitment in care of children with special needs
30	Professional Commitment and Attitude Towards improving the quality of care

Clinical training comprises all of the formal and practical "real-life" learning experiences provided for students to apply knowledge and skills in the clinical environment. Experiences would include those of short and long duration (e.g., part-time, full-time) and those that provide a variety of learning experiences (e.g., rotations on different units within the same practice setting such as Musculoskeletal Physiotherapy OPD & IPD etc.) to include comprehensivecare of patients. Each student will be under the supervision of a clinical supervisor at the clinical education site who directly instructs and supervises students during their clinical learning experiences. Clinical supervisors will evaluate each student twice during clinical training (mid- term and final) using a standard evaluation tool.

The student will be formally evaluated through **Objective Structured Clinical Examination (OSCE)** by the internal and external examiners.

Course Outcomes (COs):

At the end of the course, the students will be able to

CO1	Conduct sensitive and appropriate communication when working with children, their parents / grandparents / carers and with other health care professionals
CO2	Apply knowledge of normal development across the spectrum of childhood, & identify paediatric conditions appropriate for physiotherapy intervention.
CO3	Conduct a comprehensive paediatric physiotherapy assessment for children presenting with various conditions using appropriate outcome tools.
CO4	Conduct SMART goal setting with appropriate evidenced based interventions and educate patients and family.
CO5	Identify limits to their own practice/knowledge and recognize when to request assistance from other health professionals
CO6	Demonstrate respect for patient, parent, and family attitudes, behaviors and lifestyles and show flexibility to meet the needs of the patient and family.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3	-	-	-	-	2	3
CO2	-	3	3	-	-	-	-
CO3	-	3	3	-	-	-	-
CO4	1	-	-	3	-	2	-
CO5	-	-	-	-	3	-	3
CO6	-	-	-	-	-	2	3

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF MEDICAL SCIENCES
ASHOK & RITA PATEL INSTITUTE OF PHYSIOTHERAPY
MPT (Paediatrics) - Semester III
PT934.01-DISSERTATION I

INTRODUCTION

The dissertation is aimed to train a postgraduate student in research methods and techniques. It includes identification of a problem, formulation of a hypothesis, search and review of literature getting acquainted with recent advances, designing of a research study, collection of data, critical analysis, and comparison of results and drawing conclusions. Every candidate pursuing MPT degree course is required to carry out work on a selected research project under the guidance of a recognized postgraduate teacher. The result of such a work shall be submitted in the form of dissertation.

Course Work:

Submission of dissertation proposal to the dissertation proposal approval committee

Content of a proposal:

The proposal shall include the following sections.

1. **Introduction:** This should give a short overview of the problem, issues, or topic to be studied and reasons why it is an important problem to study. Often, it includes some historical background of the problem.
2. **Statement of the problem:** This is a more complete discussion of the problem of the dissertation. The major questions, hypotheses, or statements should be included with a possible list of secondary questions if appropriate. This section is an extended discussion of the problem often with some literature noted.
3. **Statement of the significance of this study to the field :** In other words, why is this study important? It should include a statement about how it advances knowledge in the field with reference to past literature and general concerns of the area.
4. **Literature Review related to the problem, issue, or topic:** In this, various schools of thought on this problem are explored with significant attention to the conceptual and the theoretical aspects of the problem and how those contribute to the topic's development.
5. **Methodology:** It includes a discussion of the research design, the population to be studied, discussion of appropriate instruments, sampling concerns, data collection approaches, analyses tobe conducted and a projected timetable for completion of the dissertation.
6. **Limitations:** It is to clarify or limit the scope of the method(s) employed in this study.

Once the student has received the permission from their Dissertation guide to defend their proposal, the proposal defense will be scheduled. The proposal approval committee would be constituted consisting of 3 members with one external member, one internal faculty (other than guide) from the institute and the head of institute. The dissertation committee member's backgrounds should complement the area of anticipated study. This does not require that each member be an expert on every aspect of the study. Potentially, the members will all bring different strengths or knowledge basesto the committee.

The dissertation proposal approval committee approves the dissertation proposal subject to the furtherapproval from IRB.

The Institutional Review Board Approval Process:

Researchers who use human participants in their research must follow specific guidelines as a condition for using the data provided by these persons. If the proposal involves human participants, the student will need to submit an application to the Institutional Review Board (IRB) of the Institute. In consultation with his or her Guide, the student prepares an IRB application and submits it to the institute's IRB coordinator. The IRB judge's application on issues related to protection of participants from physical and emotional distress and not on theoretical or methodological grounds. The study cannot go forward until IRB approval is received. In many cases, minor revisions to the study will be necessary to gain approval. Collecting data prior to receiving IRB approval is considered sufficient grounds for halting a research study. In general, students should not submit their IRB application before the proposal defense as any major changes in the proposal would necessitate re-applying to the IRB.

This proposal is considered an agreement or contract describing how the student will conduct thestudy and cannot be changed without guide/committee approval. Substantial changes to the methods, goals, and objectives articulated in a proposal will necessitate a new proposal defense.

Course Outcomes (COs):

At the end of the course, the students will be able to

CO1	Recognise the importance of planning and preparation required to undertake a research project
CO2	Articulate a clear research question or problem and formulate a hypothesis
CO3	Know existing body of research relevant to their topic and explain how their project fit
CO4	Critically analyse and evaluate the knowledge and understanding in relation to the agreed area of study
CO5	Design an appropriate research proposal to carry out research

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	-	2	2	2	-	-	-
CO2	2	2	2	2	-	-	-
CO3	-	-	-	2	2	-	-
CO4	-	-	-	2	2	-	-
CO5	-	-	-	-	-	2	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation "-"

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
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ASHOK & RITA PATEL INSTITUTE OF PHYSIOTHERAPY
MPT (Paediatrics) – Semester III
PT935.01 PHYSICAL & FUNCTIONAL DIAGNOSIS FOR PAEDIATRICS

Credit Hours:

Hours. / Wk.			Credits			Total Marks		Total
L	P	T	L	P	T	Theory	Practical	
3	2	5	3	1	4	100	100	200

A. Outline of the course:

Sr. No.	Title of the unit	Minimum number of hours
1.	ASSESSMENT OF TYPICAL CHILDREN	06
2.	NEURODEVELOPMENTAL ASSESSMENT	10
3	OROMOTOR ASSESSMENT	06
4	NEONATAL ASSESSMENT	05
5.	SENSORY INTEGRATION ASSESSMENT	08
6.	FITNESS TESTS	06
7.	ELECTRODIAGNOSIS	06
8.	RADIOLOGY AND OTHER INVESTIGATIONS	05
9.	MISCELLANEOUS	08

Total hours (Theory): 60

Total hours (Practical): 40

Total hours: 100

B. Detailed Syllabus:

1.	**ASSESSMENT OF TYPICAL CHILDREN**	**06 Hours**	**10%**
1.1	Factors affecting the growth of normal children		
1.2	Regional differences between Indian and other children		
1.3	Assessment: BMI & other obesity scales and questionnaires, Growth charts, Nutritional values, Qualities and components of a balanced diet,		
1.4	Nutritional disorders and other applied aspects		
2.	**NEURODEVELOPMENTAL ASSESSMENT**	**10 Hours**	**16%**
2.1	Contents, Method of administration and record maintenance		
2.2	Scales: Bayley scales of infant development (BSID-II), Bruininisk-Oseretsky Test of Motor Proficiency(BOTMP), Test of Gross Motor Development (TGMD), Peabody Developmental Motor Scale (PDMS), Gross Motor Function Measure(GMFM), Gross Motor Function Classification System(GMFCS), Denver Developmental Screening Test, Peabody Picture Vocabulary test, Stanford Binet Intelligence Scale, Wechsler Intelligence Tests, Kaufman Assessment Battery for Children, Kaufman Adolescent and Adult Intelligence test		
3.	**OROMOTOR ASSESSMENT**	**06 Hours**	**10%**
3.1	Assessment of head and neck		
3.2	Upper body posture		
3.3	Scales: Oral motor feeding rating scale, Neonatal Oral Motor Assessment Scale		
4.	**NEONATAL ASSESSMENT**	**05 Hours**	**8%**
4.1	Identifying high risk infants, HIE staging		
4.2	NICU environment		
4.3	Scales: APGAR, Brazelton Neonatal Behavioral Assessment Scale, INFANIB, Alberta infant motor scale, Test of Infant Motor Performance, Bayley scales of Infant development, Brazelton Neonatal Behavioral Assessment Scale, Dubowitz scale		
5.	**SENSORY INTEGRATION ASSESSMENT**	**08 Hours**	**14%**
5.1	Identifying the need of the sensory integration assessment		
5.2	Behavioural scales		
5.3	Comprehension scales		
5.4	Test of sensory function in infants		
5.5	Sensory integration and Praxis tests		
5.6	Miller assessment for Preschoolers		
5.7	Vineland adaptive behavior scales		

6.	**FITNESS TESTS**	**06 Hours**	**10%**
6.1 6.2	Administration and comparison with the reference values Tests: Cardiovascular endurance tests, Body composition measurement, Musculoskeletal function, Physical activity test		
7.	**ELECTRODIAGNOSIS**	**06 Hours**	**10%**
7.1	EMG-NCV interpretation for the paediatric conditions.		
8.	**RADIOLOGY AND OTHER INVESTIGATIONS**	**05 Hours**	**8%**
8.1 8.2 8.3	Procedure Need of the laboratory investigations with the reference values Interpretation: CT scan, MRI, X- ray, Carotid angiography, Myelography, Muscle and Nerve biopsy, CSF examination		
9.	**MISCELLANEOUS**	**08 Hours**	**14%**
9.1 9.2 9.3 9.4 9.5	Burns: Lund & Browder chart, Vancouver scar assessment scale Quality of life assessment scales Pain assessment scales Gait analysis Posture analysis		

Course Outcomes:

At the end of the semester the student will be able to

CO1	Demonstrate professional behaviour and respectful communication with patients and caregiver
CO2	Apply knowledge of the normal development of a child and identify deviation from the normal.
CO3	Apply advanced clinical reasoning skills and an evidence-based approach to evaluate paediatric condition
CO4	Identify outcome measures to comment on diagnosis and prognosis of paediatric conditions
CO5	Apply knowledge to determine the need for special investigation and refer the patient for the same.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3	-	-	-	-	-	-
CO2	-	2	3	2	-	-	-
CO3	-	-	3	2	-	-	-
CO4	-	3	2	-	-	-	-
CO5	-	-	-	-	3	2	2
CO6	-	-	-	-	-	-	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

Recommended Study Material:

❖ **Textbooks:**

1. Handbook of Neurologic rating scales: Robert M. Herndon
2. Electrodiagnosis in disease of nerve and muscles: Kimuraj J, F A Davis,Philadelphia
3. Physical rehabilitation by Susan B,O’ Sullivan, Thomas J. Schmitz

❖ **Reference Books:**

1. Motor control Theory and practice: Shummway-cook,& Anne
2. Motor assessment of the developing infant: Martha C. Piper & Johanna Darrah
3. Neurological assessment of the Preterm & Full-term newborn infant: Lilly Dubowitz
4. Measurement & Evaluation in Human Performance:
5. Exploring the spectrum of Autism & Pervasive developmental disorders: Carolyn Murray-Slutsky & Betty A. Paris
6. Sensory Integration with diverse populations: Susan Smith Roley.

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF MEDICAL SCIENCES
ASHOK & RITA PATEL INSTITUTE OF PHYSIOTHERAPY
MPT (Paediatrics) - Semester III
PT936.01 ADVANCED PHYSIOTHERAPEUTICS FOR PAEDIATRICS

Credit Hours:

Hrs. / Wk.			Credits			Total Marks		Total
L	P	T	L	P	T	Theory	Practical	
4	2	6	4	1	5	100	100	200

A. Outline of the course:

Sr No.	Title of the unit	Minimum No. of hours
1.	DECISION MAKING IN PEDIATRIC PHYSIOTHERAPY	7
2.	MANAGEMENT OF MUSCULOSKELETAL DISORDERS	15
3.	MANAGEMENT OF NEUROLOGICAL DISORDERS	12
4.	MANAGEMENT OF CARDIOPULMONARY DISORDERS	10
5.	BURNS	9
6.	GENETIC DISORDERS	7
7	PEDIATRIC ONCOLOGY	5
8	NEONATAL DISORDERS	15

Total hours (Theory): 80 Hrs.

Total hours (Practical): 40 Hrs.

Total hours: 120 Hrs.

B. Detailed Syllabus:

1	**DECISION MAKING IN PEDIATRIC PHYSIOTHERAPY**	**7 hrs**	**10%**
1.1	Evidence Based Decision Making		
1.2	ICF as a Framework for clinical decision making		
1.3	Child centered care		
1.4	Family centered care		
1.5	Hypothesis oriented paediatric focus algorithm – HOPFA		
2	**MANAGEMENT OF MUSCULOSKELETAL IMPAIRMENT**	**15 hrs**	**20%**
	Etiology & pathophysiology, impairments and activity limitations, Physiotherapy examination & goal setting, Physiotherapy interventions and prognosis of the following conditions:		
2.1	Muscular Dystrophies		
2.2	Limb deficiencies and amputations		
2.3	Perthe's Disease		
2.4	Sports injuries in children		
2.5	Juvenile Rheumatoid Arthritis		
2.6	Congenital Muscular Torticollis		
2.7	Spinal disorder (Scoliosis, Kyphoscoliosis)		
2.8	Arthrogryposis Multiplex Congenita		
2.9	Osteogenesis imperfect		
3	**MANAGEMENT OF NEUROLOGICAL IMPAIRMENT**	**12 hrs**	**15%**
	Etiology & pathophysiology, impairments and activity limitations, Medical & Surgical management, Physiotherapy examination & goal setting, Physiotherapy interventions and prognosis of the following conditions:		
3.1	Cerebral palsy		
3.2	Developmental Coordination Disorders		
3.3	Brachial plexus injury		
3.4	Brain and Spine vascular disease - Moya Moya disease		
3.5	Spina Bifida		
3.6	Behavioural Disorders (ADHD, ASD)		
3.7	Paediatric Headache (Brain Tumour, Meningitis)		

4	**MANAGEMENT OF CARDIOPULMONARY MPAIRMENT**	**10 hrs**	**13%**
	Etiology & pathophysiology, impairments and activity limitations, Physiotherapy examination & goal setting, interventions and prognosis of:		
4.1	Cystic fibrosis		
4.2	Asthma		
4.3	Congenital heart disorders: Cyanotic & Acyanotic.		
4.4	Diaphragmatic dysfunctions		
4.5	Respiratory distress syndrome		
4.6	Aspiration syndromes		
5	**BURNS**	**9 hrs**	**10%**
5.1	Classification of burns		
5.2	Impairment and Complications		
5.3	Burns unit and team care		
5.4	Physiotherapy examination & goal setting		
5.5	Physiotherapy interventions and prognosis		
6	**GENETIC DISORDERS**	**7 hrs**	**9%**
	Etiology & pathophysiology, impairments and activity limitations, Physiotherapy examination & goal setting, Physiotherapy interventions and prognosis of:		
6.1	Down syndrome		
6.2	Prader Willi syndrome		
6.3	Cri du chat syndrome		
6.4	Klinefelter syndrome		
7	**PEDIATRIC ONCOLOGY**	**5 hrs**	**7%**
	Etiology & pathophysiology, impairments and activity limitations, Physiotherapy examination & goal setting, Physiotherapy interventions and prognosis of different types of pediatric cancer.		
8	**NEONATAL DISORDERS**	**13 hrs**	**16%**
8.1	Conditions requiring individualized special care		
8.2	Causes and Problems of Prematurity		
8.3	Theoretical perspective for developmentally supportive care		
8.4	NICU Environment, Early intervention and its implications		
8.5	Other developmental interventions		
8.6	Identifying high risk infant and High Risk Infant Follow-up		

Course Outcomes (COs):

At the end of the semester the student will be able:

CO1	Demonstrate professionalism and respectful communication with the parents and other caretakers while informing their role in the rehabilitation of the child
CO2	Demonstrate competency of comprehensive paediatric physiotherapy assessment for children presenting with various conditions using appropriate outcome tools.
CO3	Demonstrate competency to differentially diagnose the disorders on the basis of assessment and clinical presentation of the symptoms
CO4	Develop realistic SMART goals and evidence based management plans
CO5	Apply the details of physical findings to justify the reasons for evaluation, and justify the diagnostic and therapeutic plan.
CO6	Follow the ethical guidelines and maintain confidentiality of patient information

Course Articulation Matrix

	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3	-	-	-	-	2	1
CO2	-	3	3	-	-	-	-
CO3	-	3	3	-	-	-	-
CO4	-	-	2	3	-	-	-
CO5	2	-	-	-	3	-	-
CO6	-	-	-	-	-	3	3

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation "-"

Recommended Study Materials:

❖ **Textbooks:**

1. Neurological Rehabiliation: Umphred, Darcy, A.
2. Physical therapy for Children: Suzann K. Campbell
3. Pediatric physical therapy: Jan S. Tecklin

❖ **Reference Books:**

1. Decision making in Pediatric Neurologic Physical therapy: Suzann K. Campbell
2. Developmental disabilities in Infancy and Childhood: Arnold J. Capute & Pasquale J. Accardo

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF MEDICAL SCIENCES
ASHOK & RITA PATEL INSTITUTE OF PHYSIOTHERAPY
MPT (Paediatrics) - Semester IV
PT937.01- CLINICAL EDUCATION IV

Credit Hours

Teaching Scheme				Examination Scheme				
Contact Hrs.			Credit	Theory		Practical		Total
Theory	Practical	Total		Internal	External	Internal	External	
-	10	10	5	-	-	30	70	**100**

Total Hours: 200 Hours

A. Outline of the course:

Sl. No.	Skill Sets
1	Communication and motivation skills for relatives, care givers and members of the multidisciplinary team
2	Communication skills in children of all ages with a range of complex conditions/disabilities to work with treatment programs
3	Communication skills with highly challenging conditions
4	Parental education and communication skills
5	Procedural skills and Knowledge about common outcome measures in paediatrics
6	Fine and gross motor evaluation in children
7	Sensory and motor assessment in children
8	Primitive reflex testing
9	GMFM, GMFCS scoring and interpretation
10	Diagnostic skills and knowledge
11	Common investigative procedures in children
12	Mental and emotional support skills of children and families
13	Rational for the prescription of specialized device and recommend purchase
14	Evaluation for the need of home environment and School modification
15	Interpreting medical, educational and psychological assessments
16	Assessment and Evaluation Skills to record prognosis or progression

17	Problem focused management skills
18	Exercise, Play and recreational activities to optimize the function of child with special needs
19	Standardized Guidelines, research findings and government guidelines in service provision
20	Evidence based practice to the development and improvement of clinical practice
21	Effective family and patient education.
22	Education for exercise adherence and home compliance
23	Case presentation of physical findings, evaluation, and diagnosis, patient education and therapeutic plan
24	Working effectively as a member of the rehabilitation team (Orthotics, podiatrics, speech & language therapist)
25	Evaluation for the need of referral services
26	Recognize own practice/knowledge and to request assistance from other health professionals if needed
27	Knowledge of legislative acts and child protection laws, sensitive issues such as child abuse
28	Knowledge about the governmental policies and guidelines for people with disability
29	Knowledge of ethical and medical-legal consideration in the care of children with physical limitations
30	Care, compassion and commitment in care of children with special needs
31	Professional Commitment and Attitude Towards improving the quality of care
32	Being a responsible healthcare team member
33	Clinical governance, ability to audit own and teams practice

Clinical training comprises all of the formal and practical "real-life" learning experiences provided for students to apply knowledge and skills in the clinical environment. Experiences would include those of short and long duration (e.g., part-time, full-time) and those that provide a variety of learning experiences (e.g., rotations on different units within the same practice setting such as Paediatric Physiotherapy OPD, Paediatric OPD & IPD, NICU, PICU etc.) to include comprehensive care of patients. Each student will be under the supervision of a clinical supervisor at the clinical education site who directly instructs and supervises students during their clinical learning experiences. Clinical supervisors will evaluate each student twice during clinical training (mid-term and final) using a standard evaluation tool.

The student will be formally evaluated through **Objective Structured Clinical Examination (OSCE)** by the by internal and external examiners.

Course Outcomes (COs):

At the end of the course, the students will be able to

CO1	Demonstrate appropriate communication with attention to the different age groups: including when to address questions to child versus parent.
CO2	Conduct comprehensive & prioritized systematic assessments for specific paediatric conditions using appropriate assessment tool
CO3	Apply scientific method and critically evaluate the literature, assimilate new information, and apply this knowledge to goal setting & patient care
CO4	Demonstrate excellence in clinical decision making effectively, and efficiently educates patient and family.
CO5	Demonstrate accountability to patients, families and the health care team.
CO6	Demonstrate responsibilities as a healthcare team member and effectively draw on system resources to provide care that is of optimal value.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3	-	-	-	-	2	3
CO2	-	3	3	-	-	-	-
CO3	-	3	3	-	-	-	-
CO4	1	-	-	3	-	2	-
CO5	-	-	-	-	3	-	3
CO6	-	-	-	-	-	2	3

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF MEDICAL SCIENCES
ASHOK & RITA PATEL INSTITUTE OF PHYSIOTHERAPY
MPT (Paediatrics) - Semester IV
PT938.01- DISSERTATION II

Course work:

Submission of the completed dissertation

Content :

Every candidate shall submit to the Registrar (Academic) of the university in the prescribed proforma, within 1 month from the date of commencement of the course on or before the dates notified by the university.

The dissertation should be written under the following headings.

1. Introduction
2. Aims or objectives of study
3. Review of literature
4. Methodology
5. Results
6. Discussion
7. Conclusion
8. Summary
9. References
10. Tables & Figures
11. Annexure.

The printed text of dissertation should not be less than 50 pages/2500 words and shall not exceed 75 pages excluding references, tables, questionnaires and other annexure. It should be neatly typed in double line spacing (Font 12, Times New Roman) on one side of paper (A4 Size, 8.27” X 11.69”) and Hard bound properly (No Spiral binding). Four copies of dissertation thus prepared shall be submittedto the Registrar (Evaluation), one month before final examination on or before the dates notified by the university duly certified by the guide, head of the department and head of the institution. In theDissertation the Candidate should not disclose his Identity or of the Guide or Institution in anyway.

Three examiners appointed by the university shall evaluate the dissertation. Two of the examiners (external) shall be from outside university and one examiner (internal) will be from the institute.

A candidate who has submitted his/her dissertation once is not required to submit a fresh dissertationif he/she reappears for the examination in the same branch on the subsequent occasion, provided the dissertation has been accepted by the examiners.

If the student has submitted her examination form & also his/her dissertation previously, he/she will be permitted to give the examination within a period of 4 years anytime in future provided the dissertation has been accepted. The terms satisfactorily kept by him will be valid for a period of 4 years subsequent to submission of the dissertation after which he/she will have to undergo Post-graduate training again for terms to be eligible for appearing for theory & Practical examination.

POST-GRADUATE GUIDE:

A PG guide must have a Post-Graduate Degree in Physiotherapy with at-least 5 years of full timeteaching in the core areas of physiotherapy after post-graduation. Notwithstanding the above clause, in a case of acute shortage of qualified Post-Graduate guides, a PG teacher with 3 years full time teaching experience after Masters Degree can be considered. The PG guide can only guide student belonging to his/her specialized branch. These clauses may be reviewed after two years. The age of teacher /guide shall not exceed 62 years and the guide student ratio shall be 1: 3.

Co-guide: may be included provided the work requires substantial contribution from a sister departmentor from another medical institution recognized for teaching /training by the University. The co- guide shall be a recognized postgraduate teacher of the University

Change of Guide: In the event of a recognized guide leaving the college for any reason or in the event of death of guide, another recognized guide may take over the duties of the guide with prior permission from the university.

Course Outcomes (COs):

At the end of the course, the students will be able to

CO1	Identify, analyse and interpret suitable data to enable the research question to be answered
CO2	Demonstrate the ability to collate and critically assess/interpret data
CO3	Develop an ability to effectively communicate knowledge in a scientific manner
CO4	Describe the process of carrying out independent research in written format and report your results and conclusions with reference to existing literature
CO5	Work autonomously in an effective manner, setting and meeting deadlines
CO6	Present the research effectively in a conference setting or a written publication

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	-	-	2	2	2	-	1
CO2	1	-	3	1	-	1	1
CO3	3	1	2	-	-	-	1
CO4	1	-	3	-	-	1	-
CO5	-	-	1	1	3	2	3
CO6	3	-	-	1	1	3	3

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation "-"